Updates of B mesons' Studies in LHC Run 3

Novel probes of the Quark-Gluon Plasma medium

Camilo Torres, Henrique Legoinha, Lekai yao, Nuno Leonardo,

Zhen hu, Jhovanny Andres Mejia Guisao

2026.6.19

Comments/Questions from last meeting

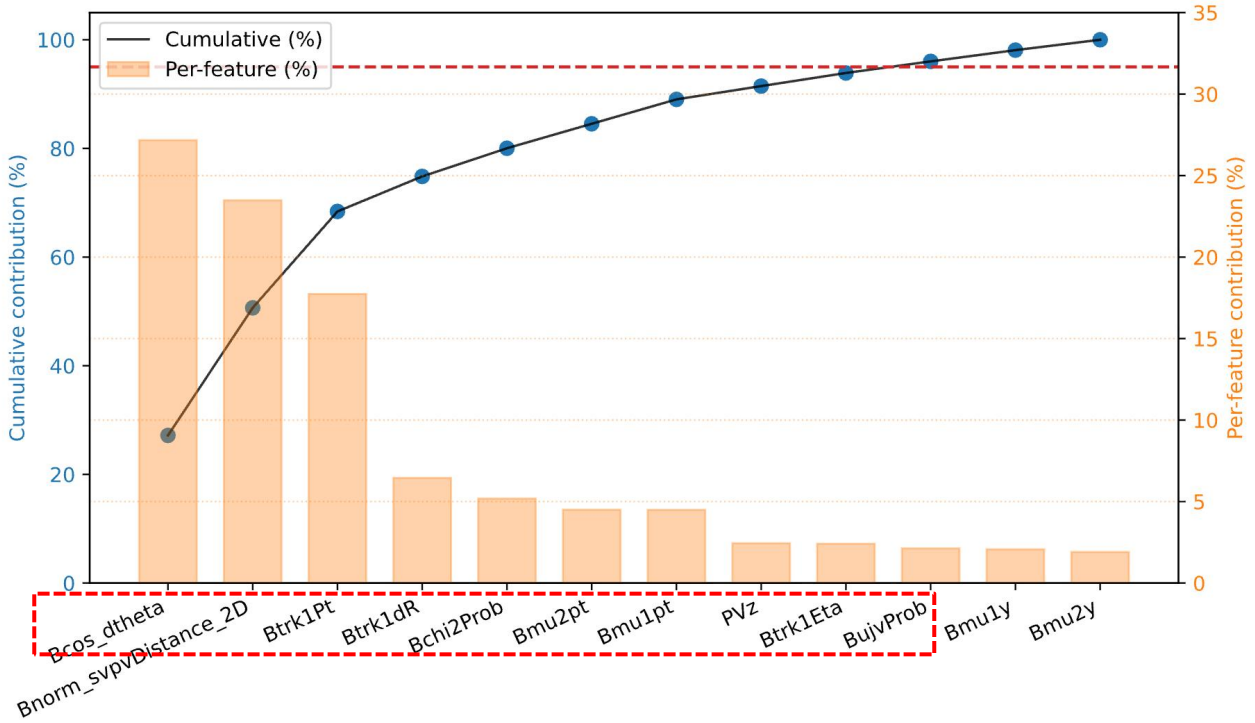
- Don’t use both BtrkdR variables in channels with di-tracks, since they can create sculpting because they are correlated with the mass
- Use strong preselection cuts in PbPb to reduce the sample size
- Bs and Bd have the contamination problem on the left of the signal

2024 pp : $B^+ \rightarrow JK^+$

12 variables used in XGBoost training:

Btrk1dR, Btrk1Pt, Bcos_dtheta, Bnorm_svpvDistance_2D, Bchi2Prob, PVz, BujvProb, Btrk1Eta, Bmu1pt, Bmu2pt(, Bmu2y, Bmu1y)

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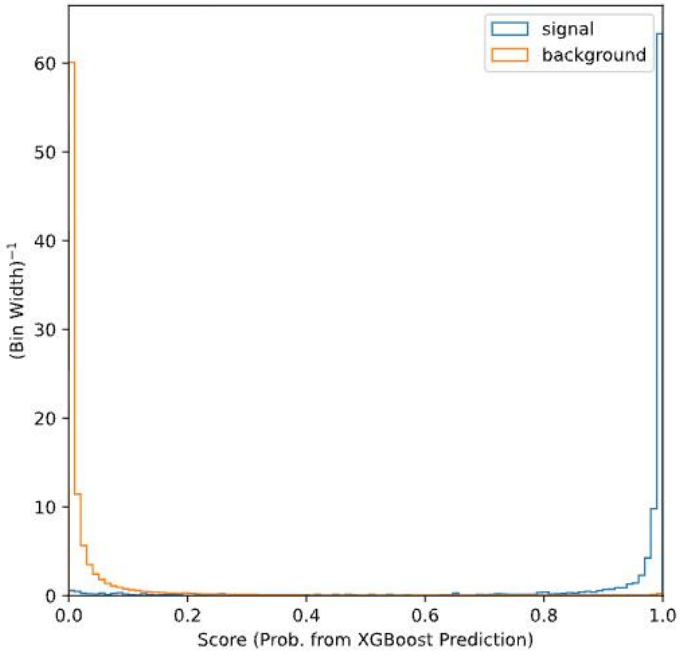
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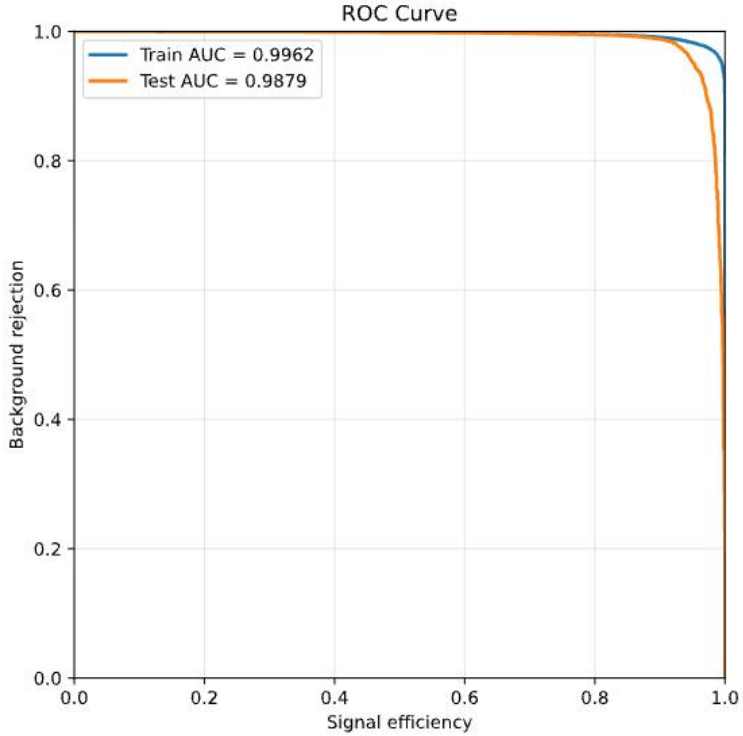
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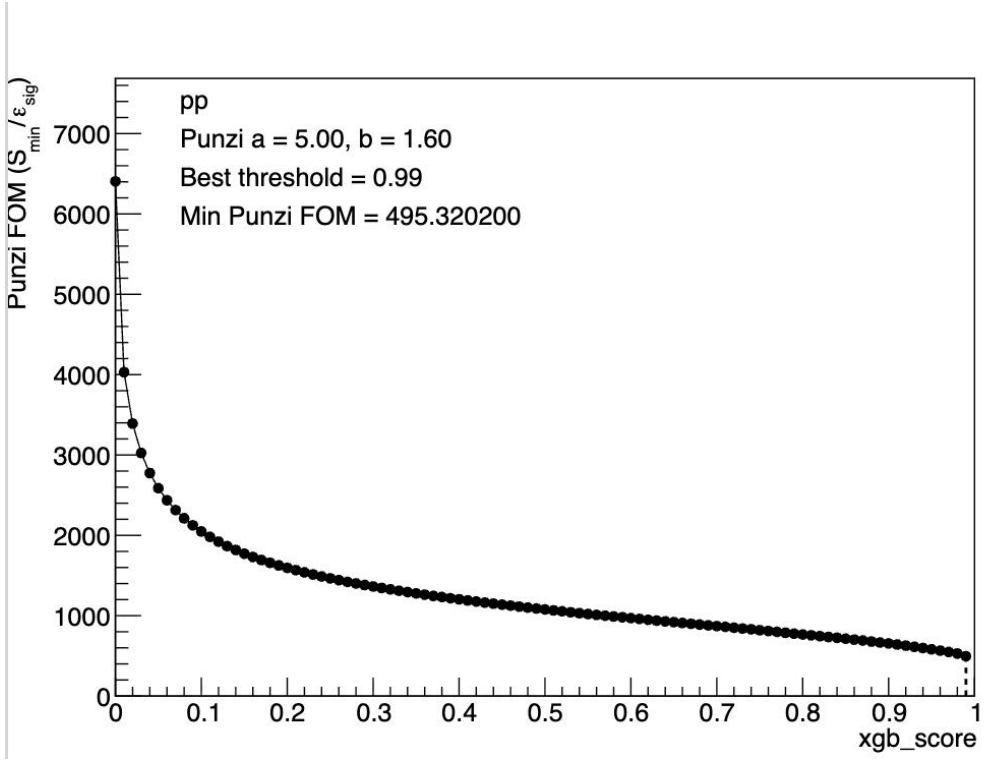
sideband region:

((5.0 < Bmass < 5.2) or (5.36 < Bmass < 5.56))

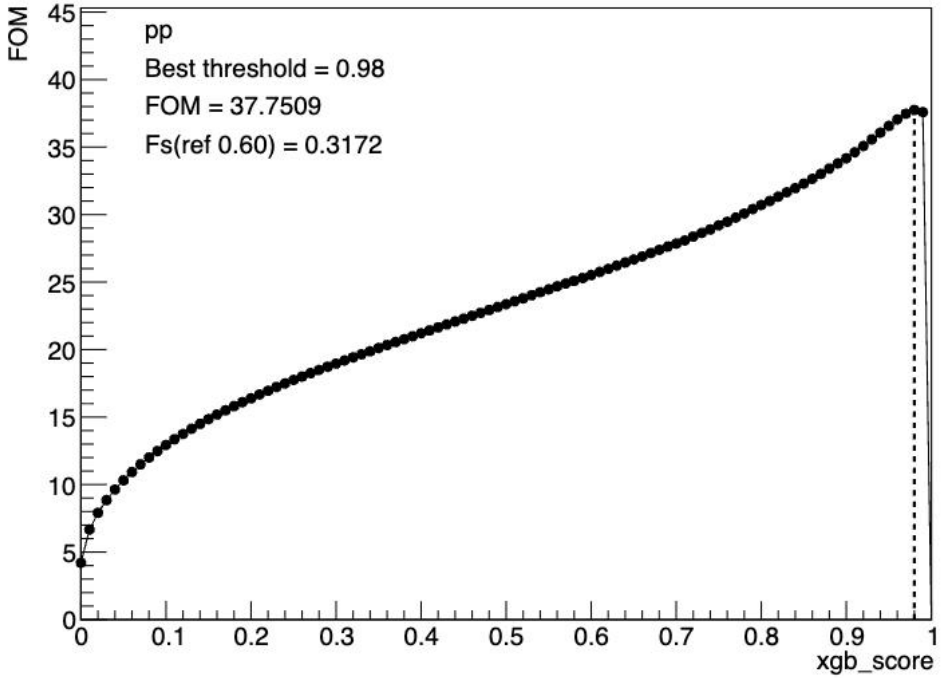




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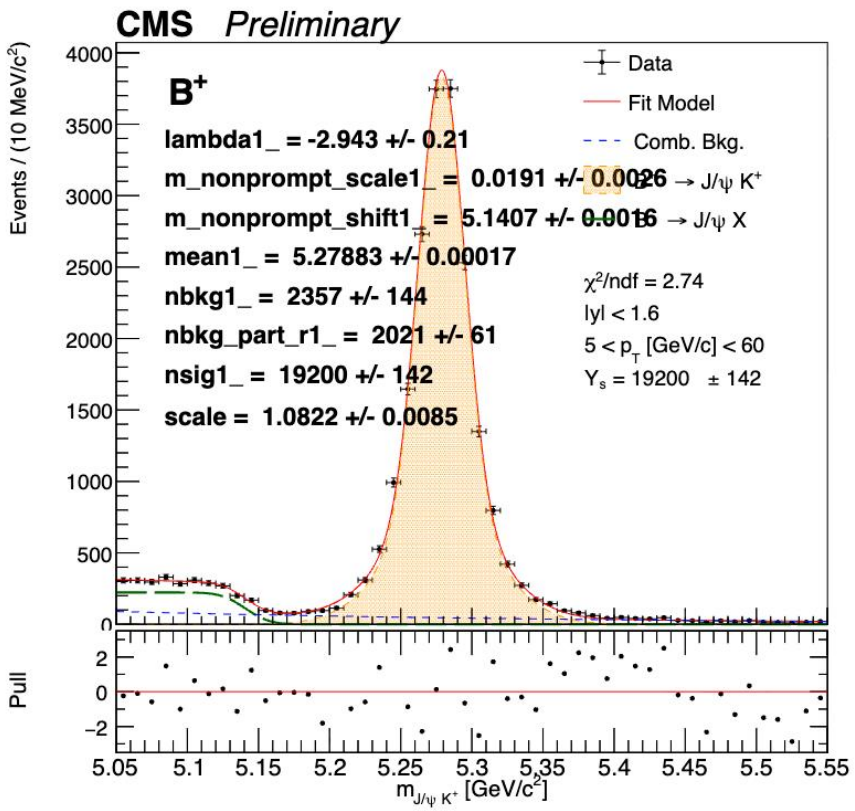


Punzi FOM

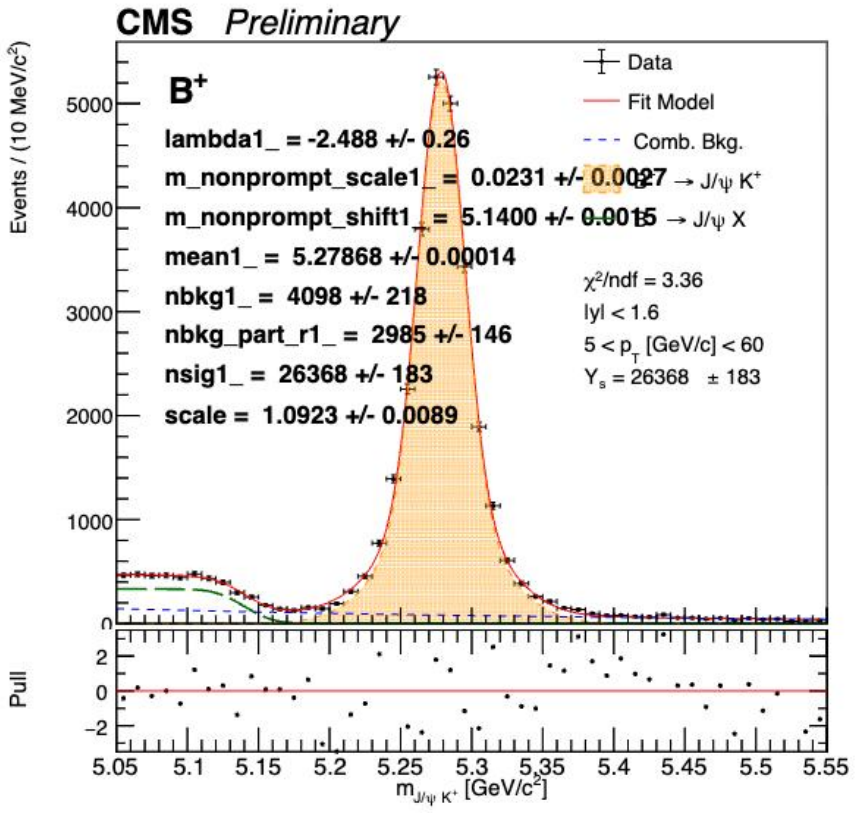


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2024 pp : $B^+ \rightarrow J K^+$



with punzi optimal cut



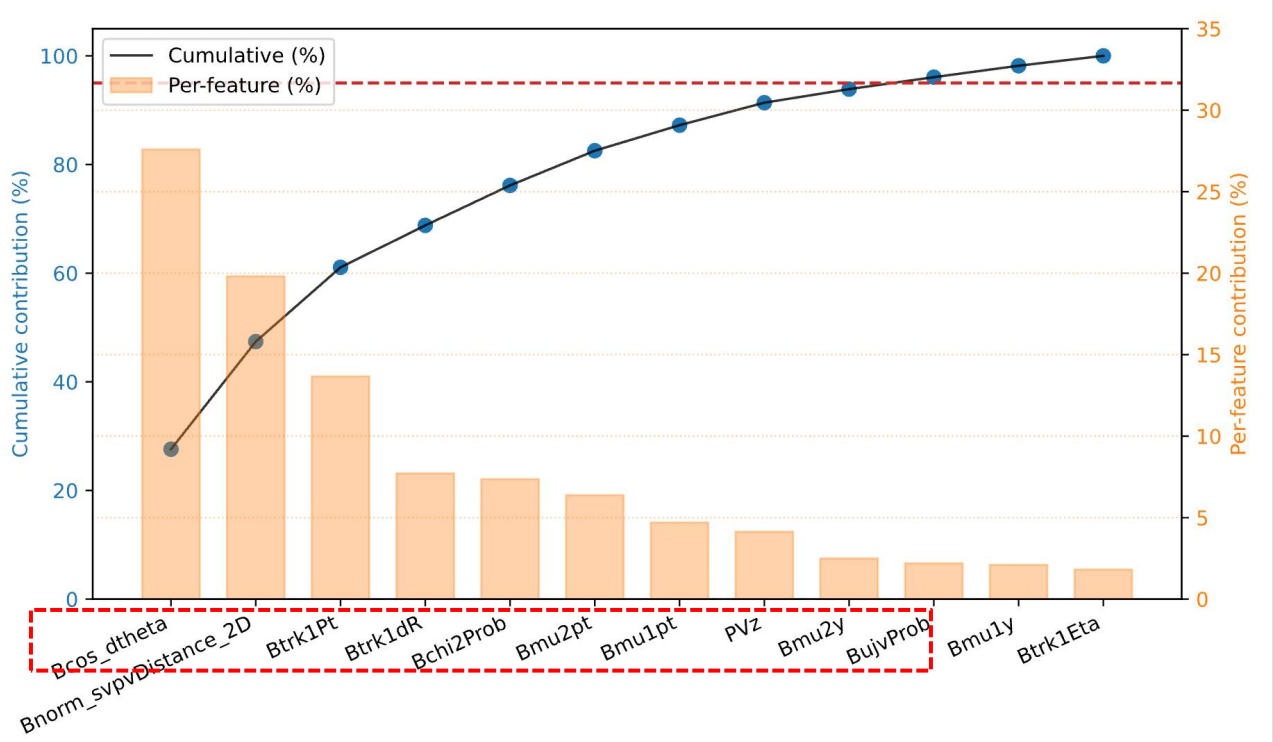
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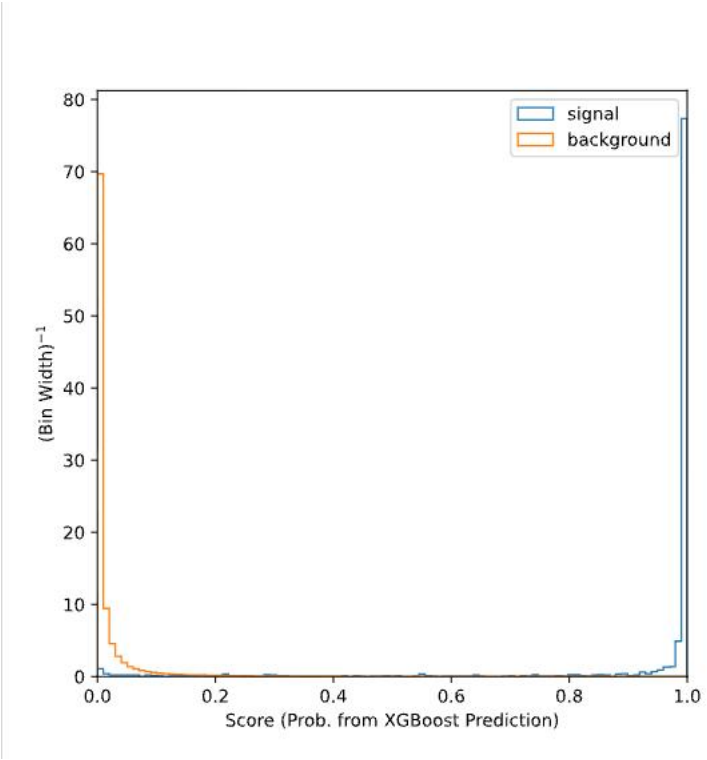
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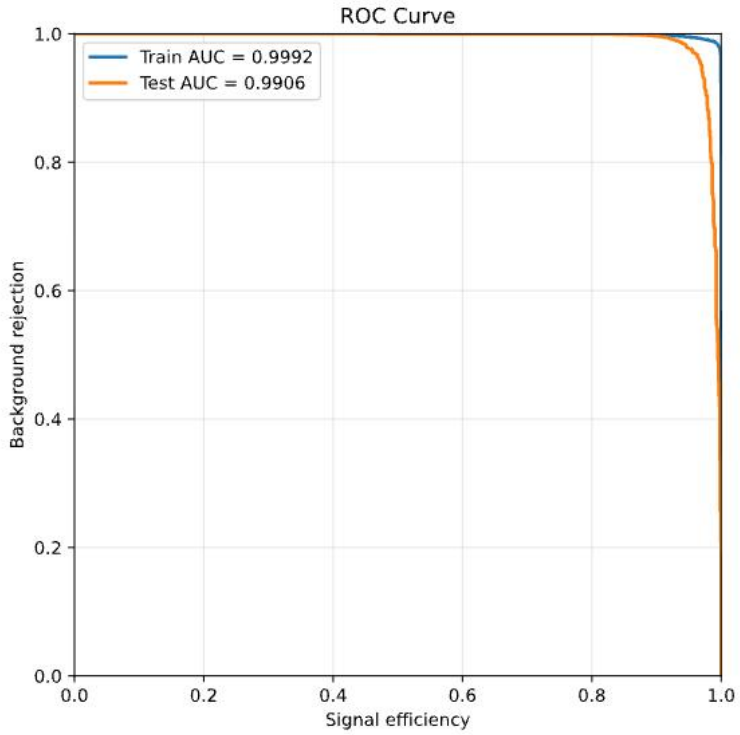
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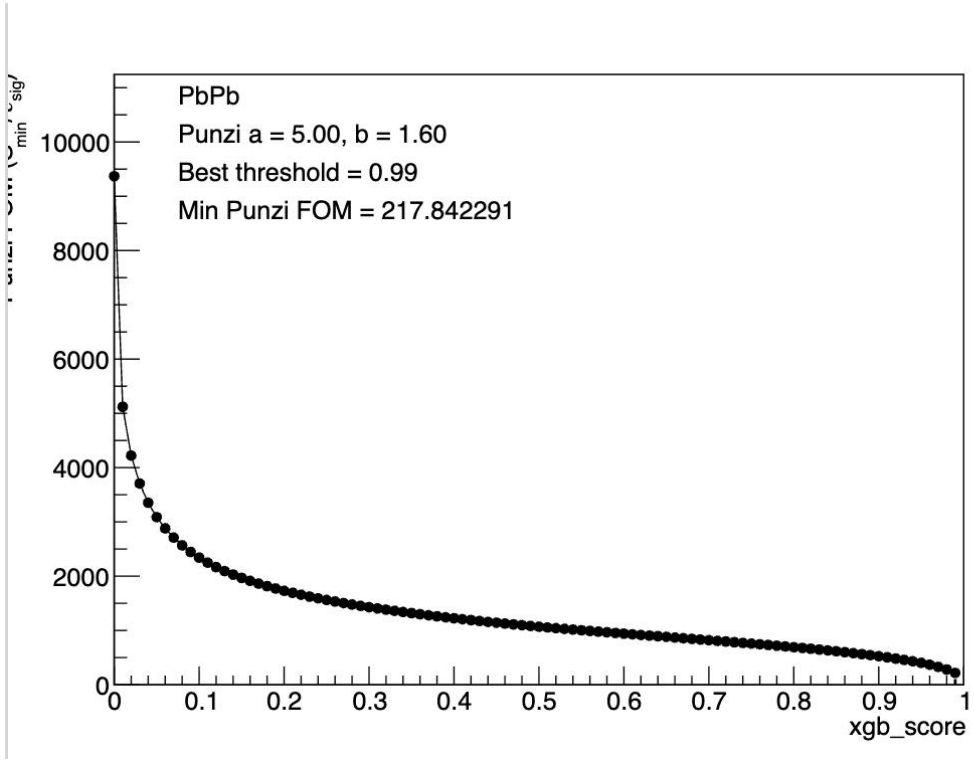
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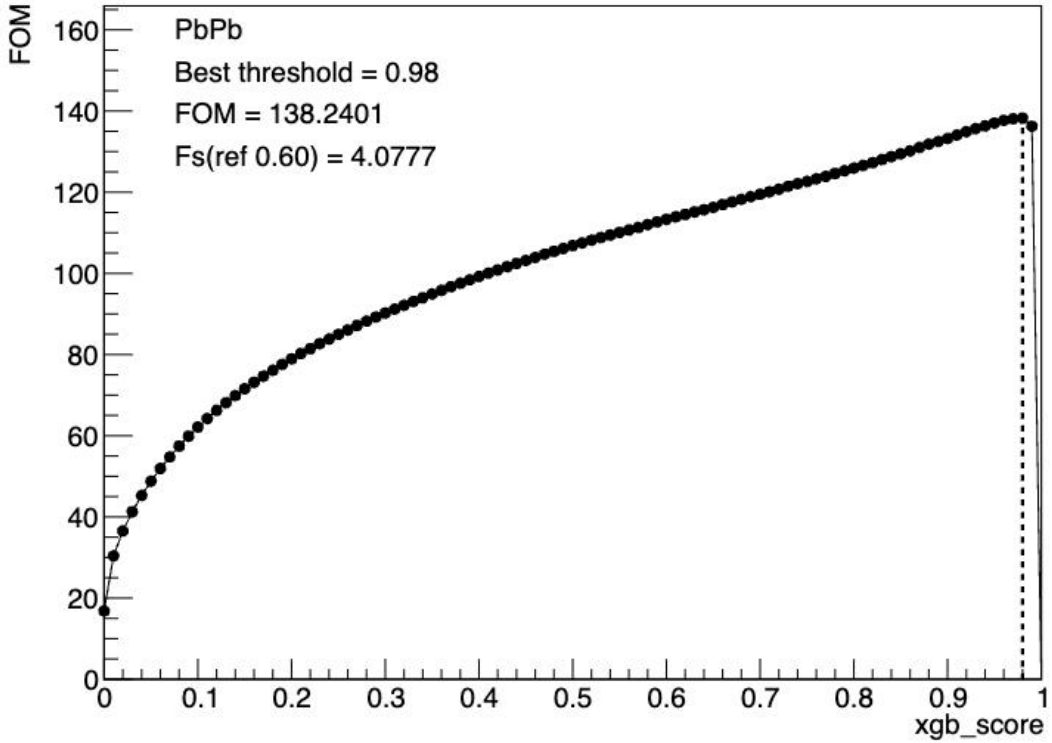




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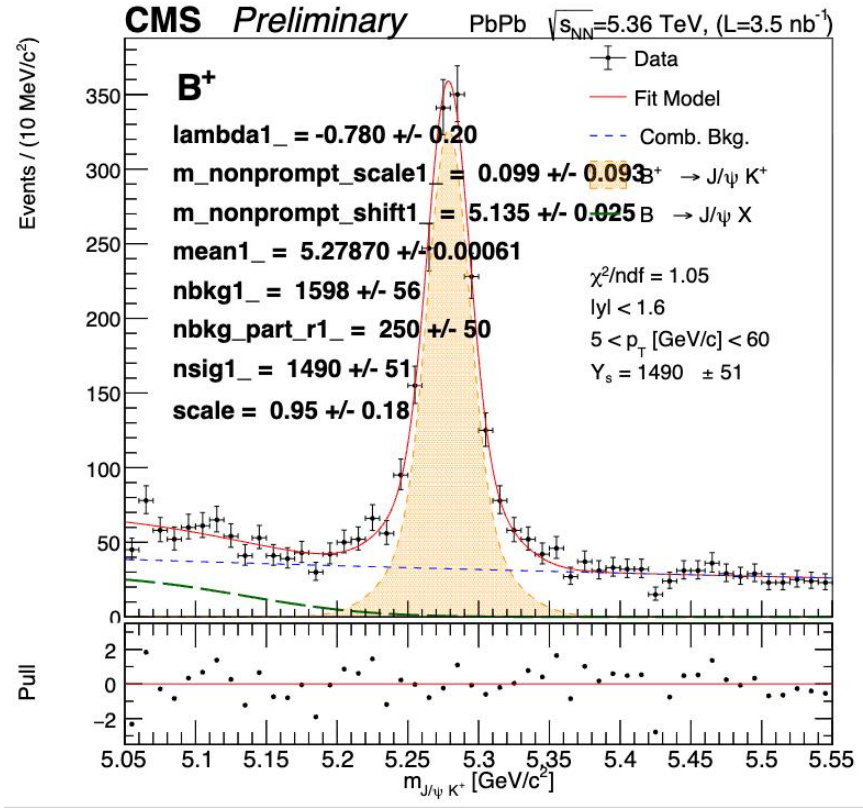


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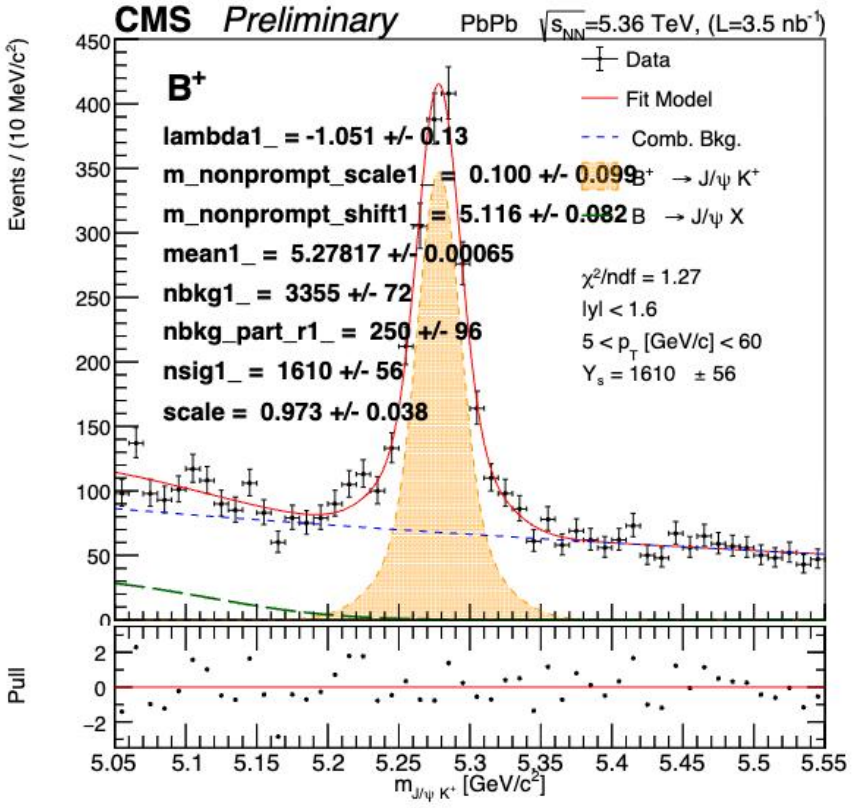


FOM

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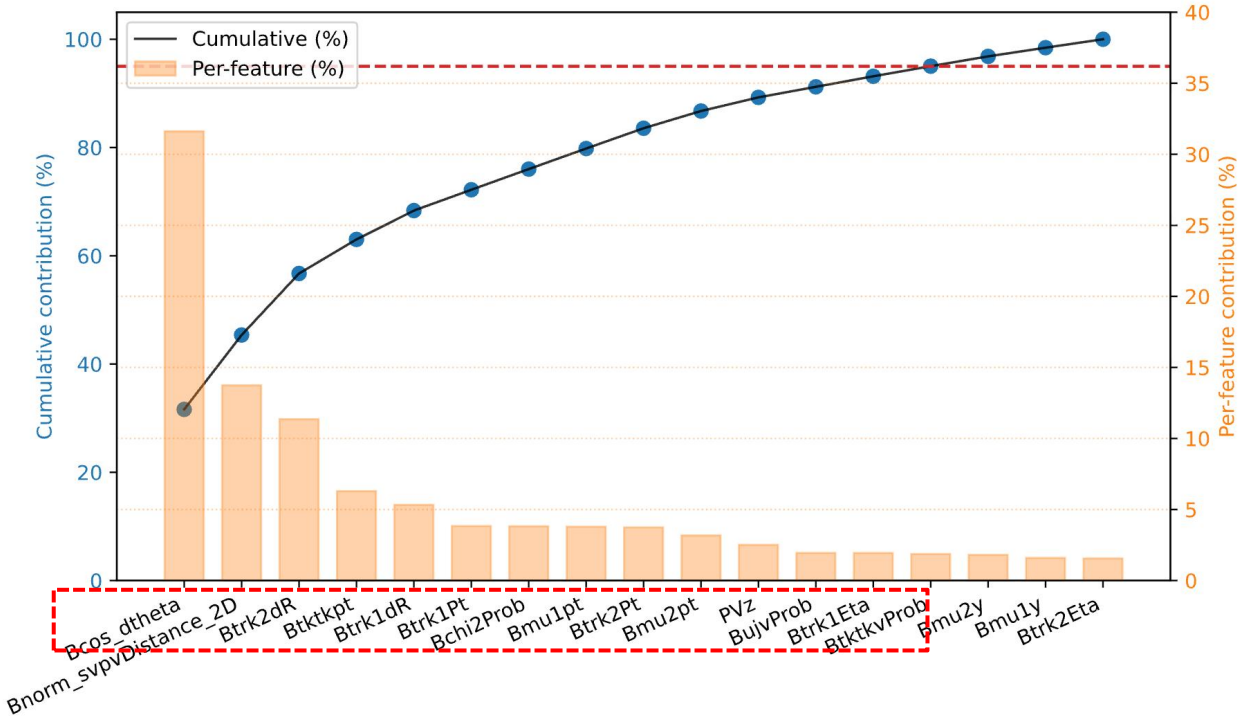
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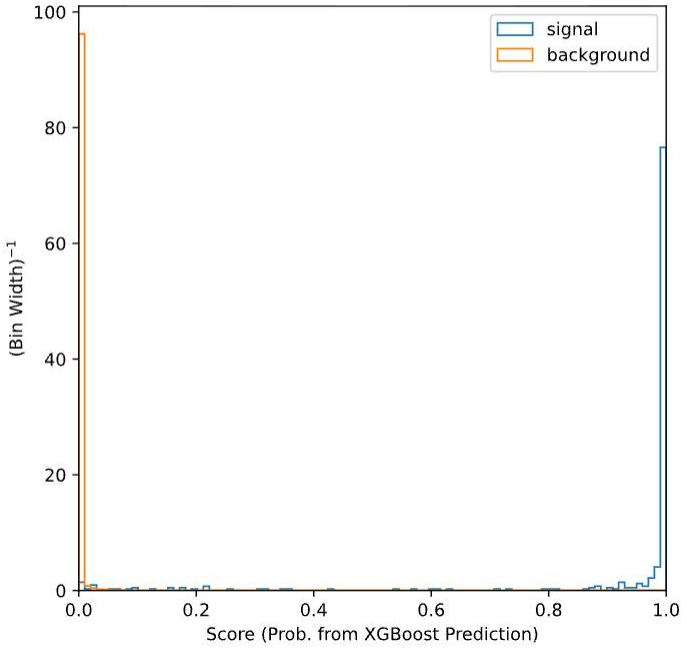
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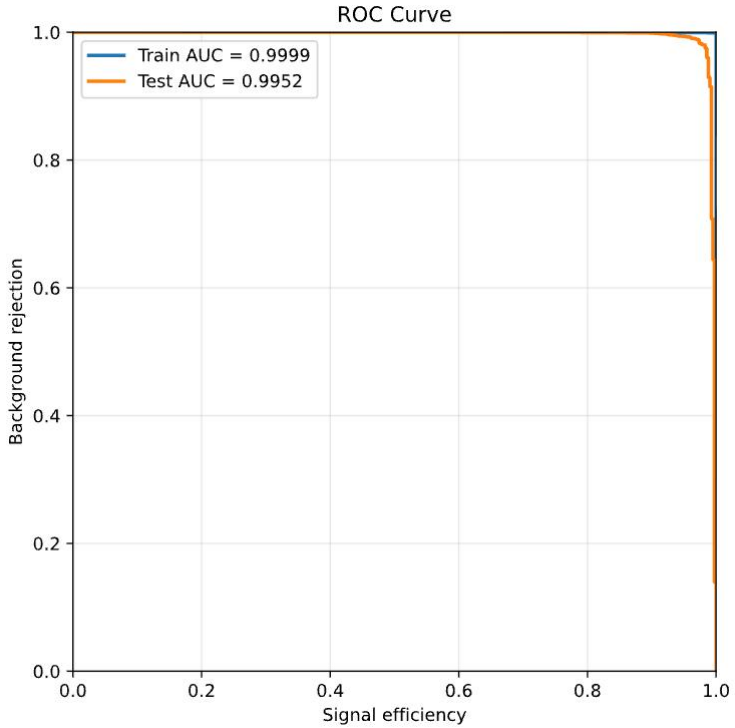
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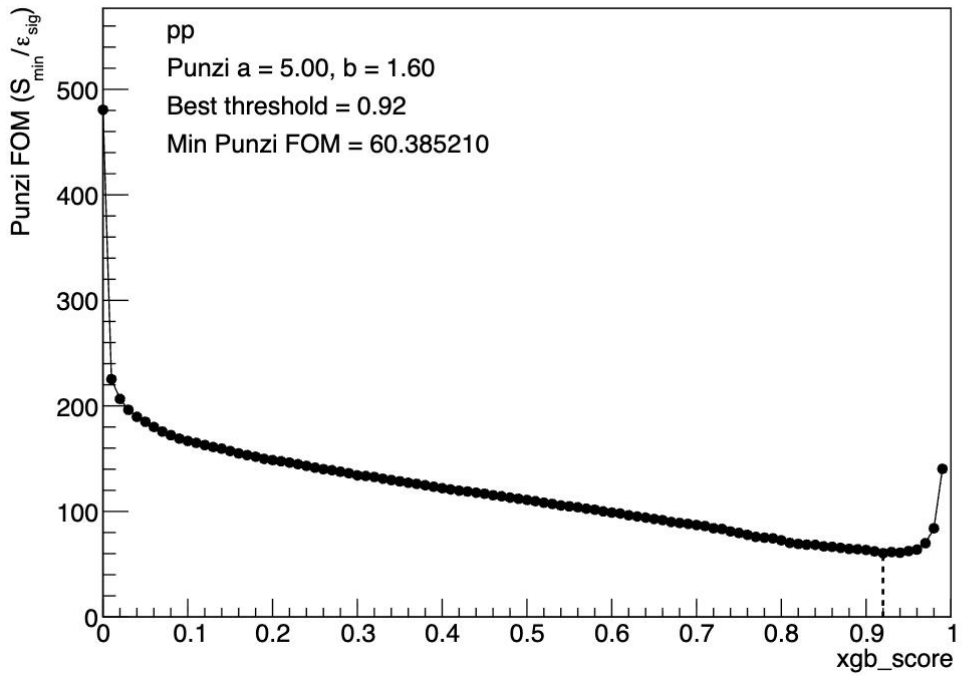
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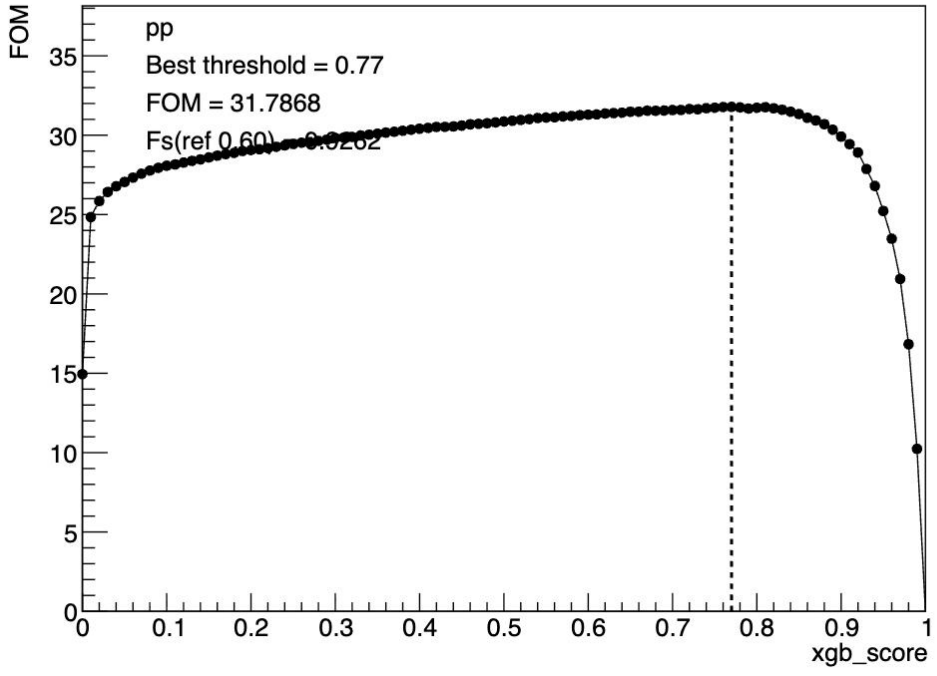




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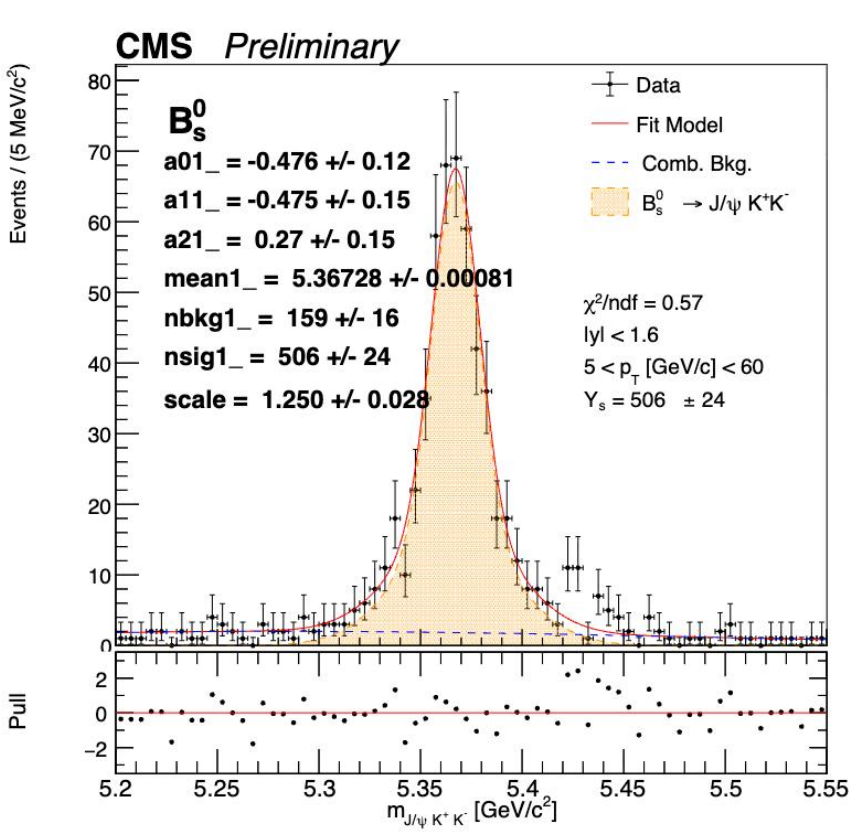


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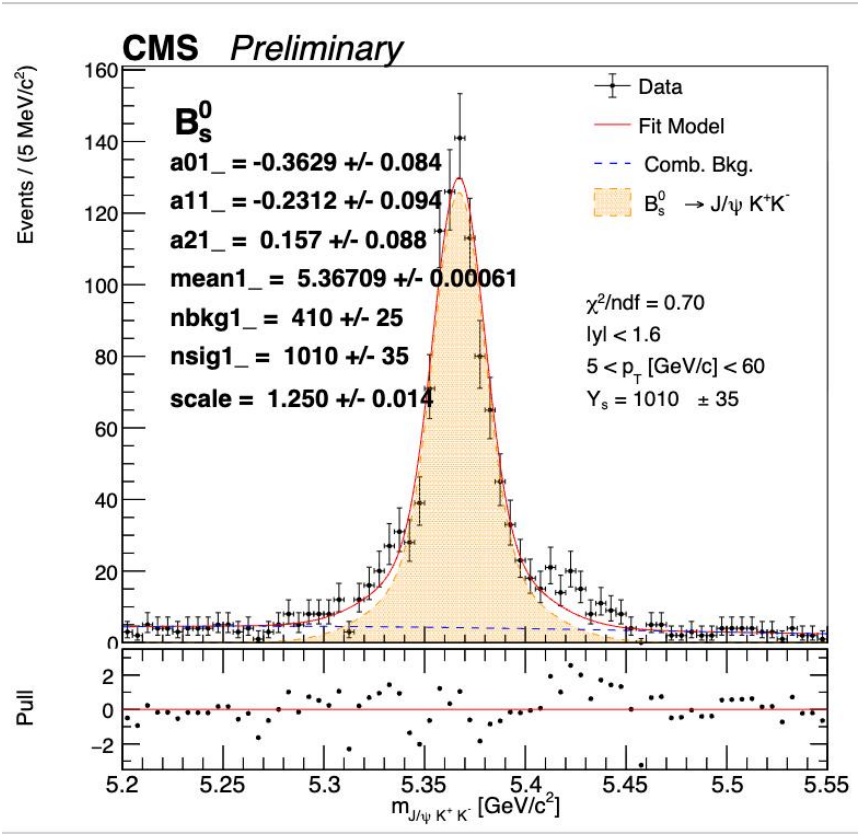


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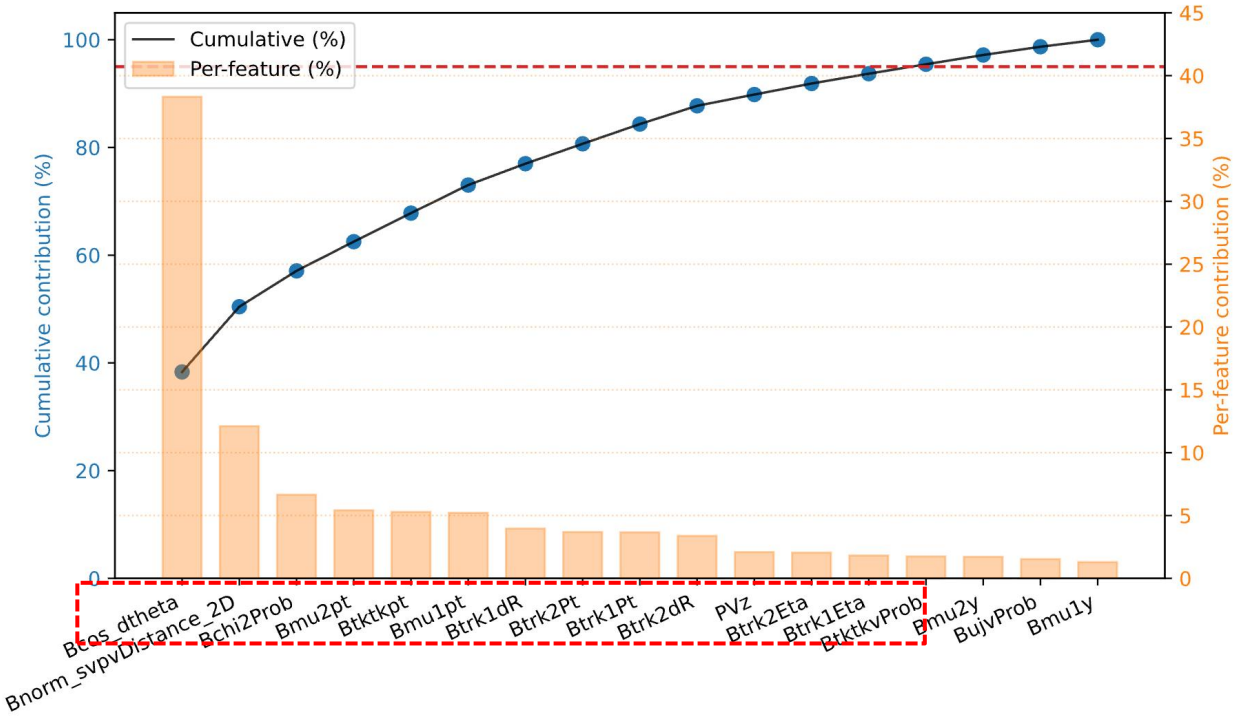
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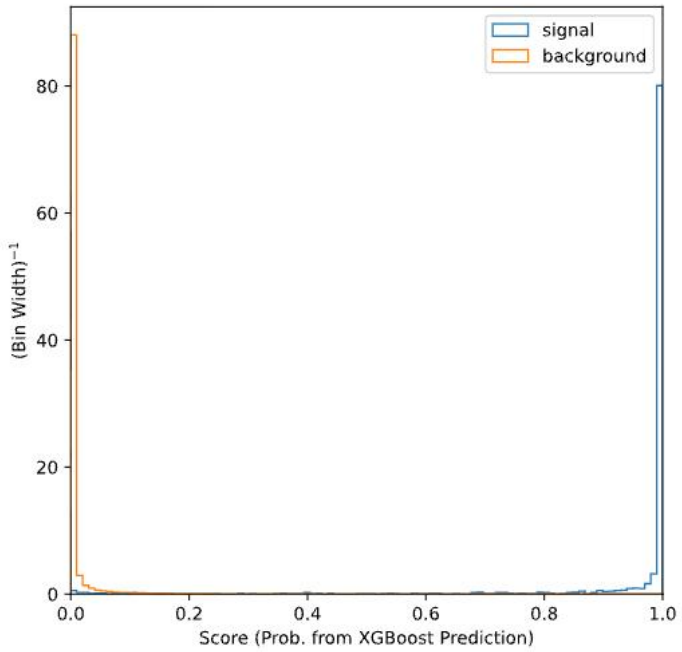
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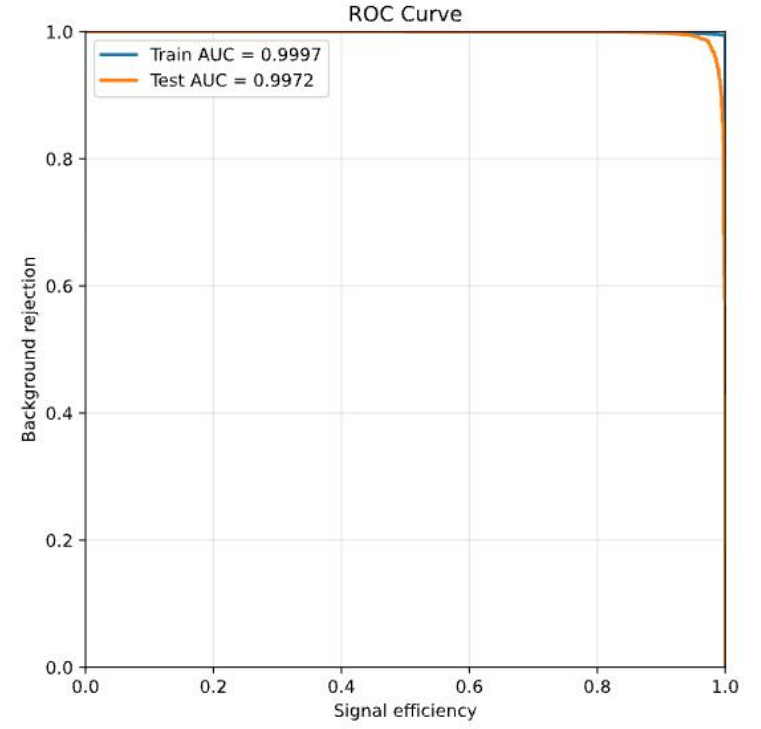
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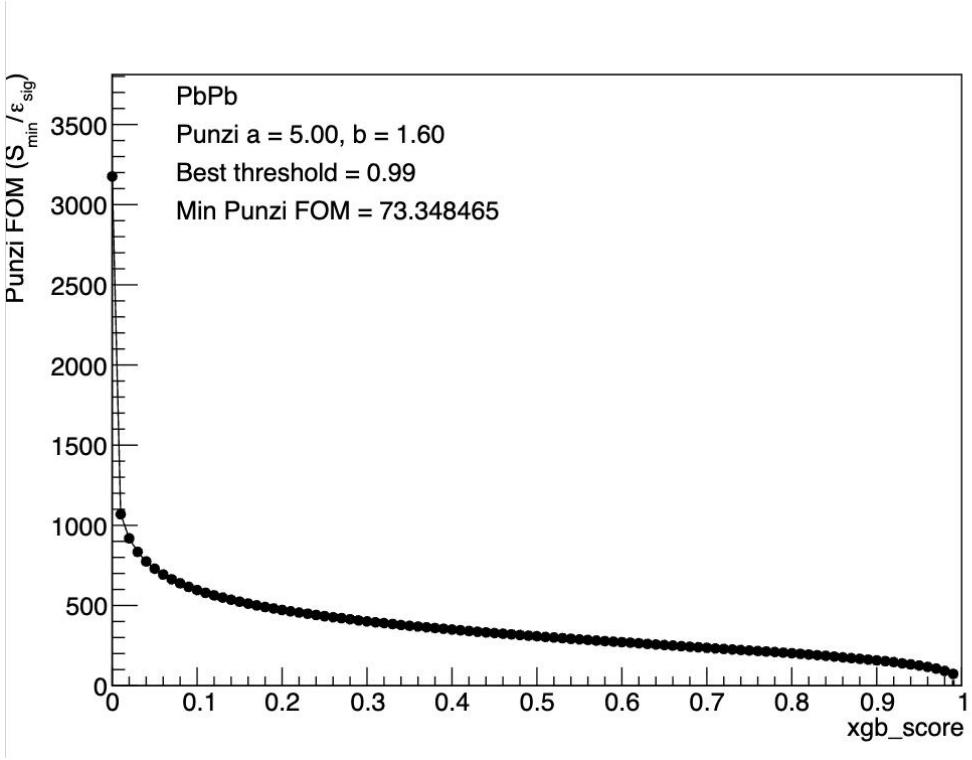
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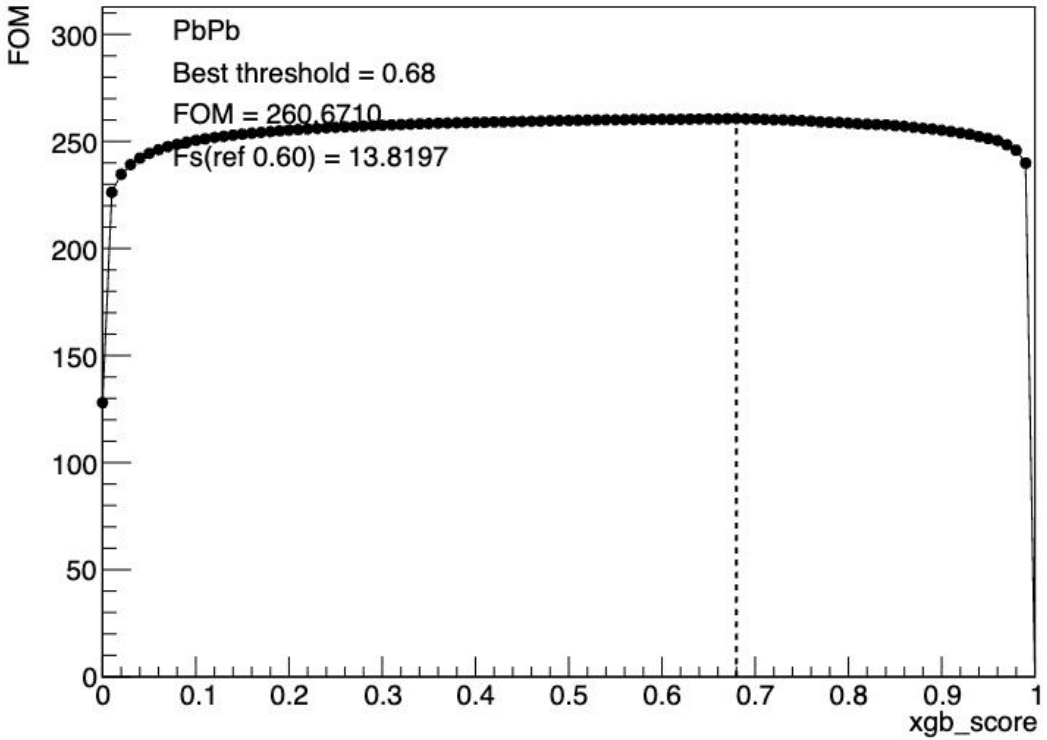




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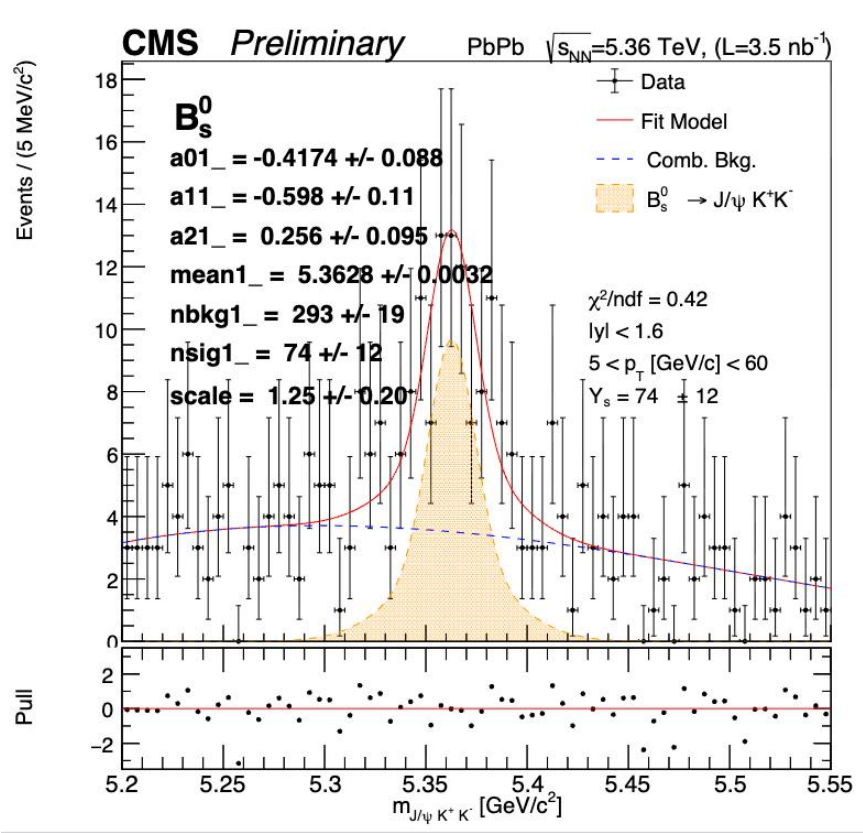


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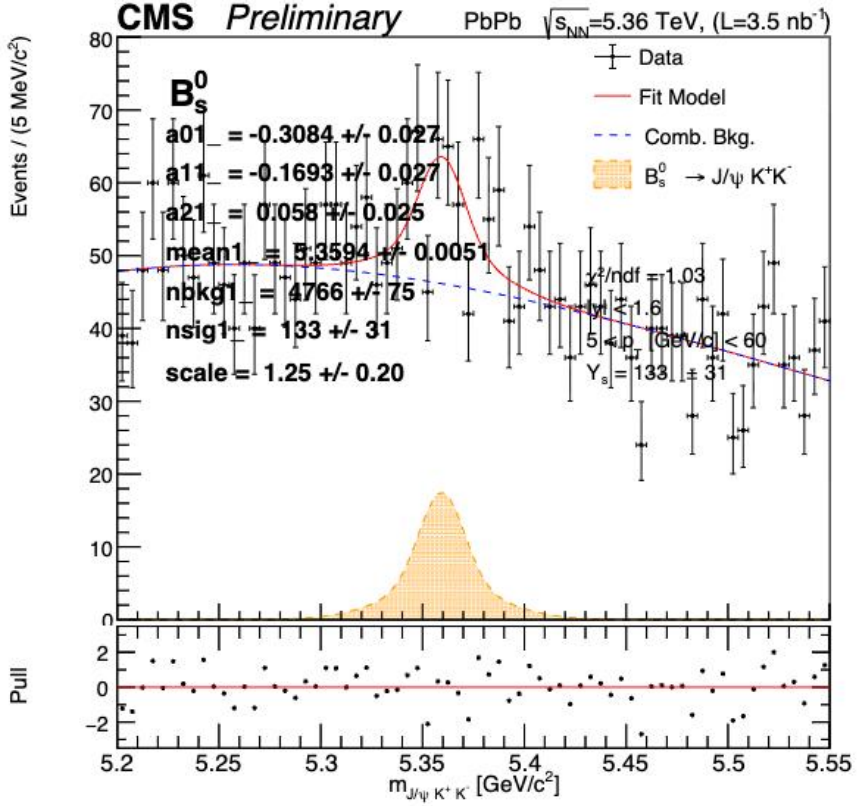


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with punzi optimal cut



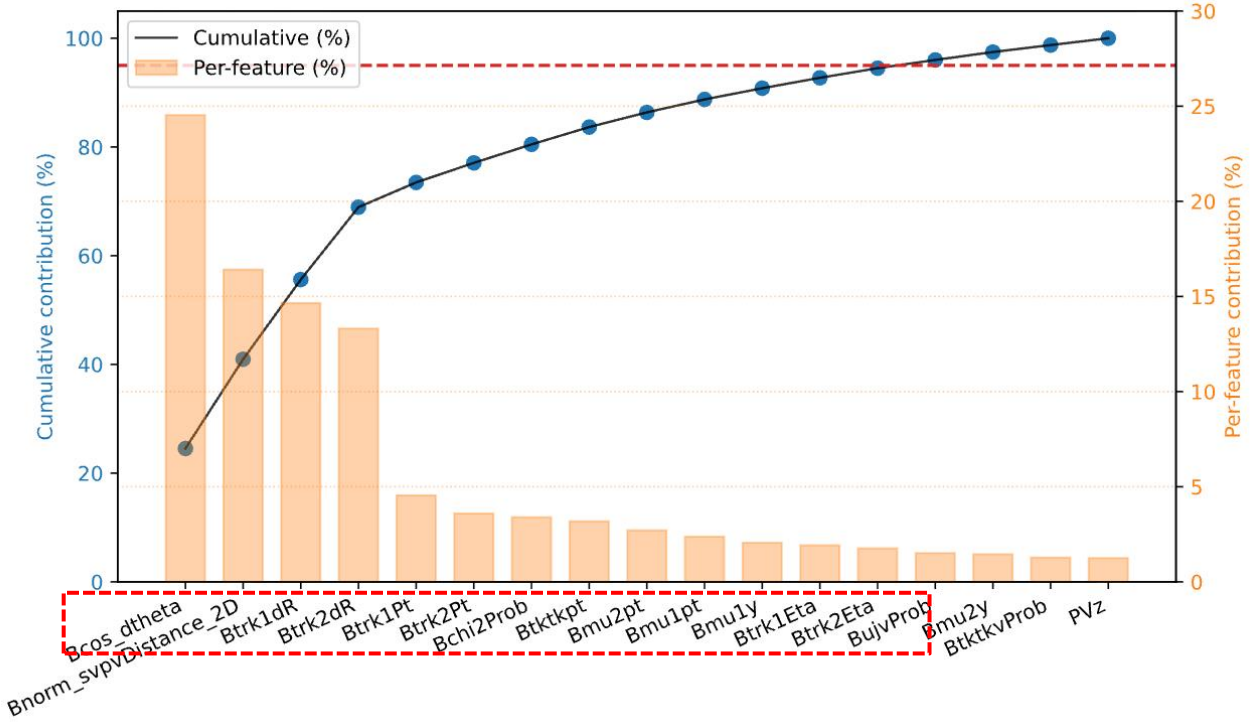
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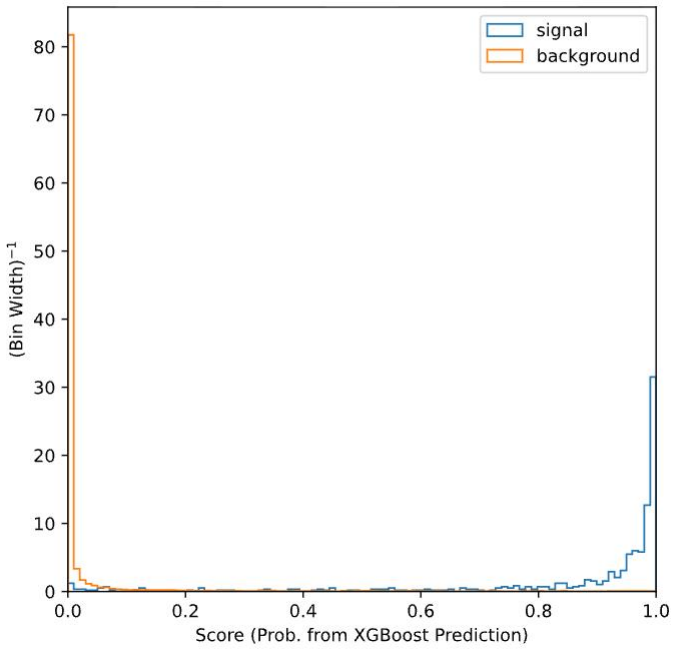
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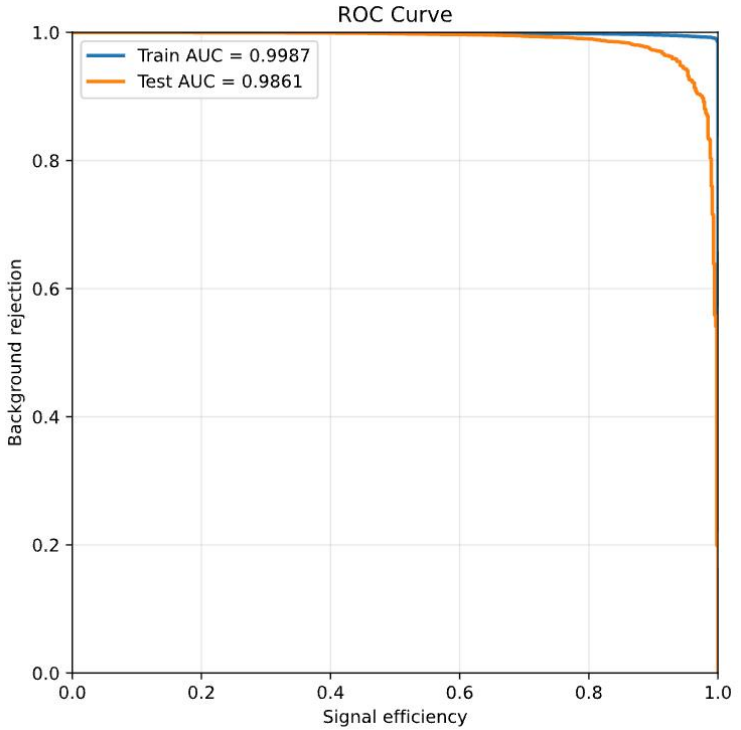
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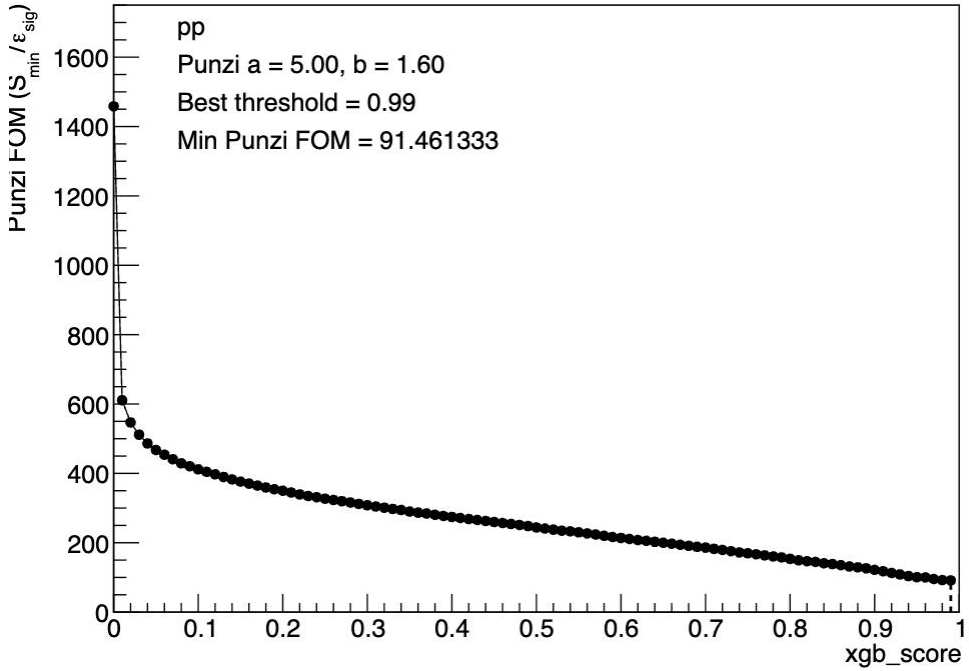
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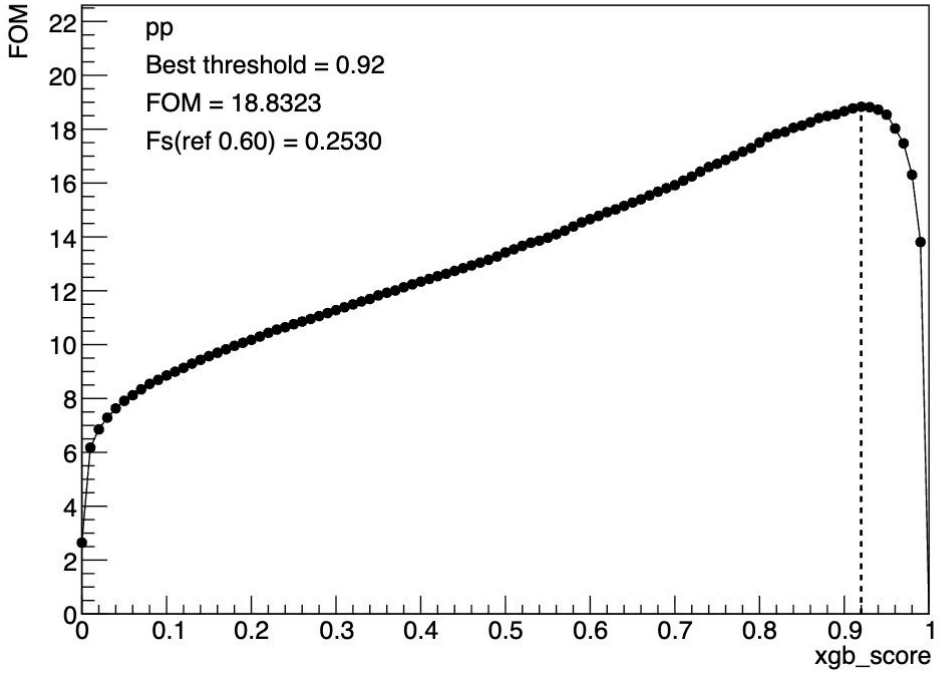




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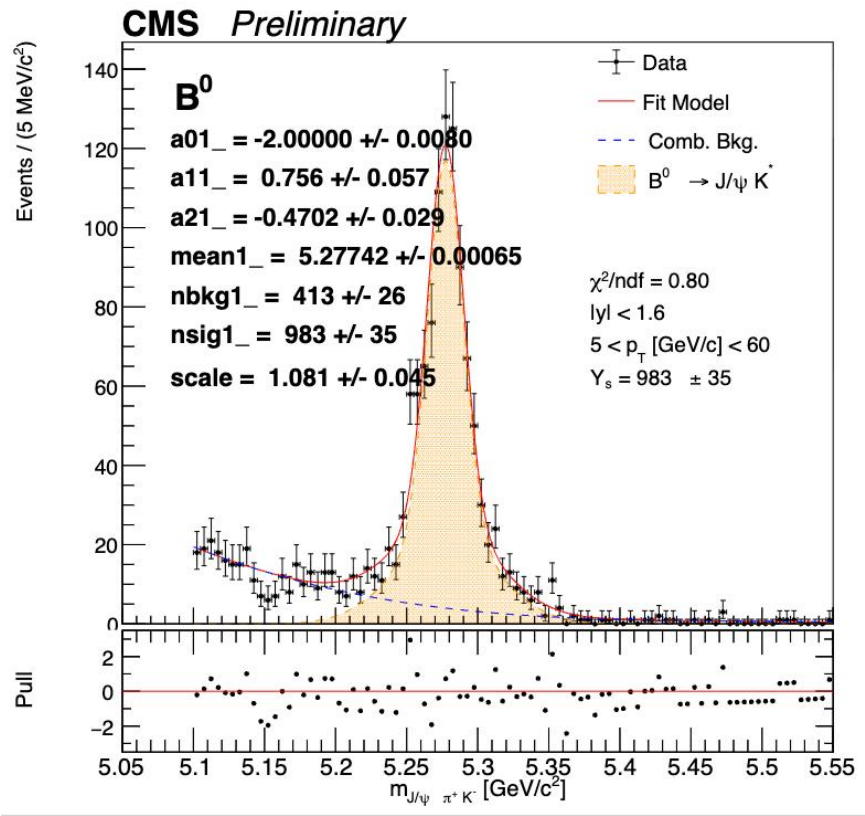


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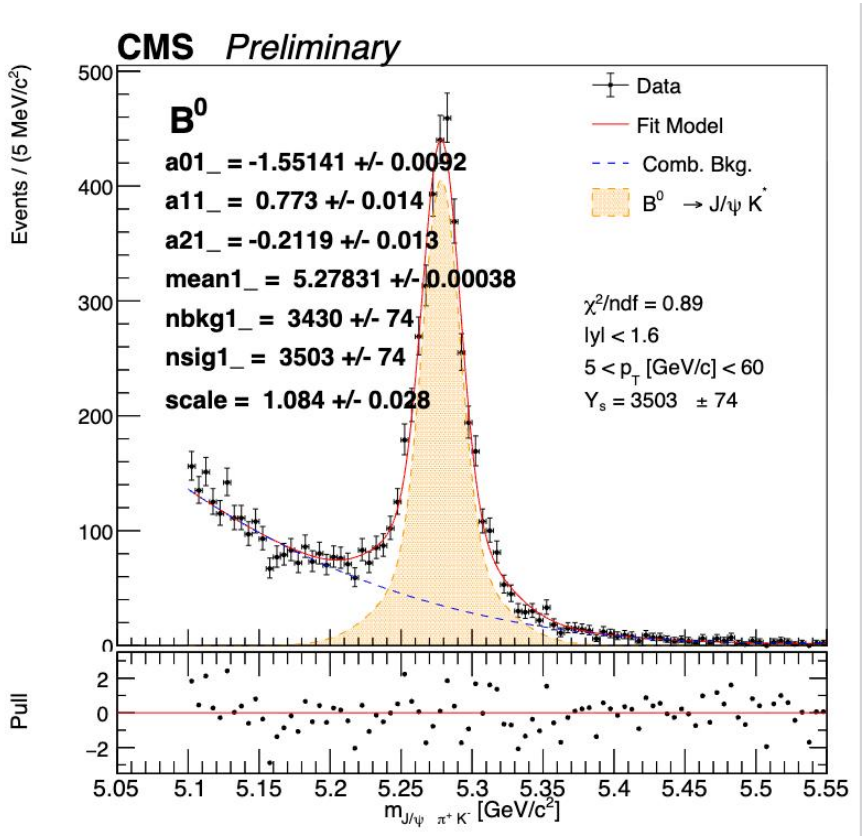


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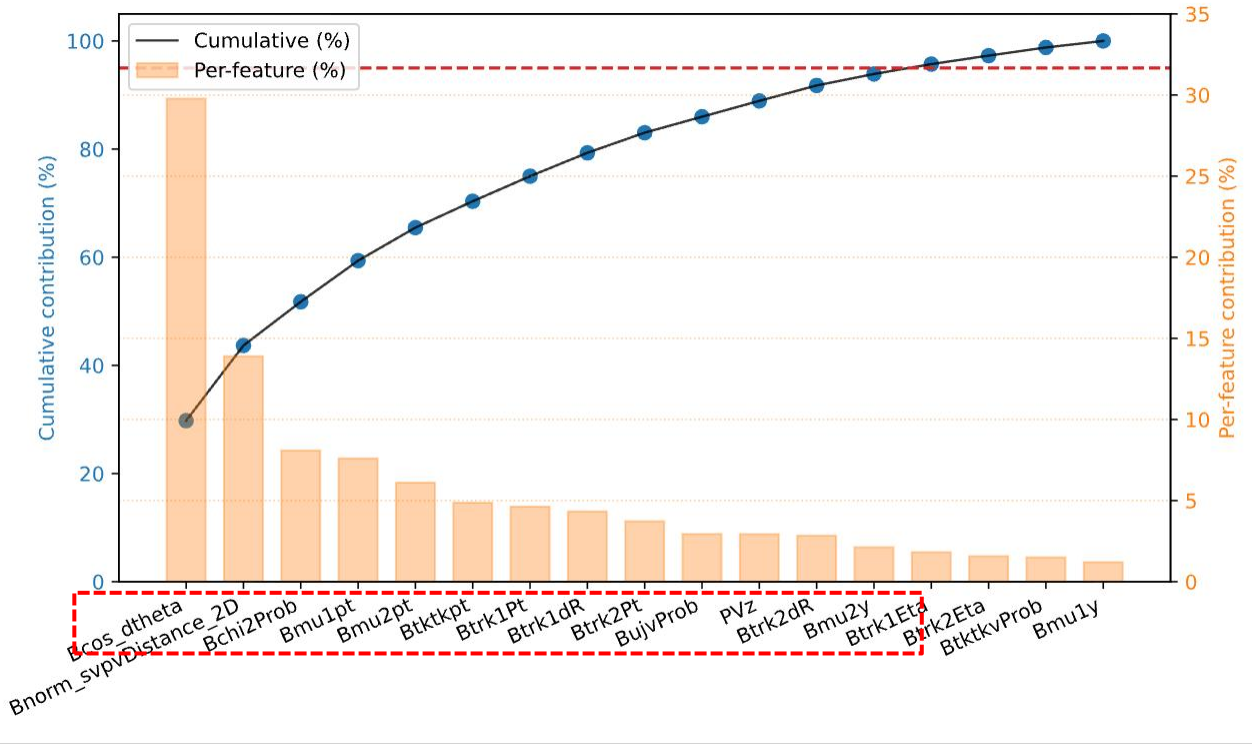
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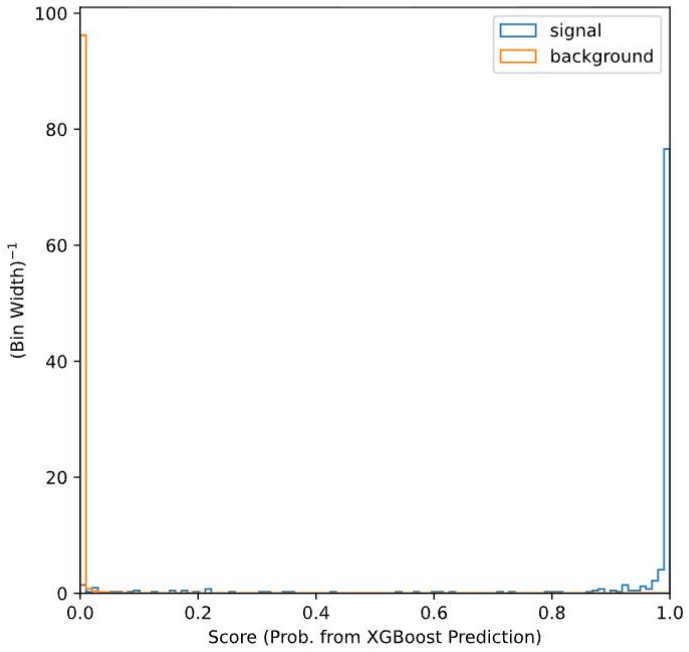
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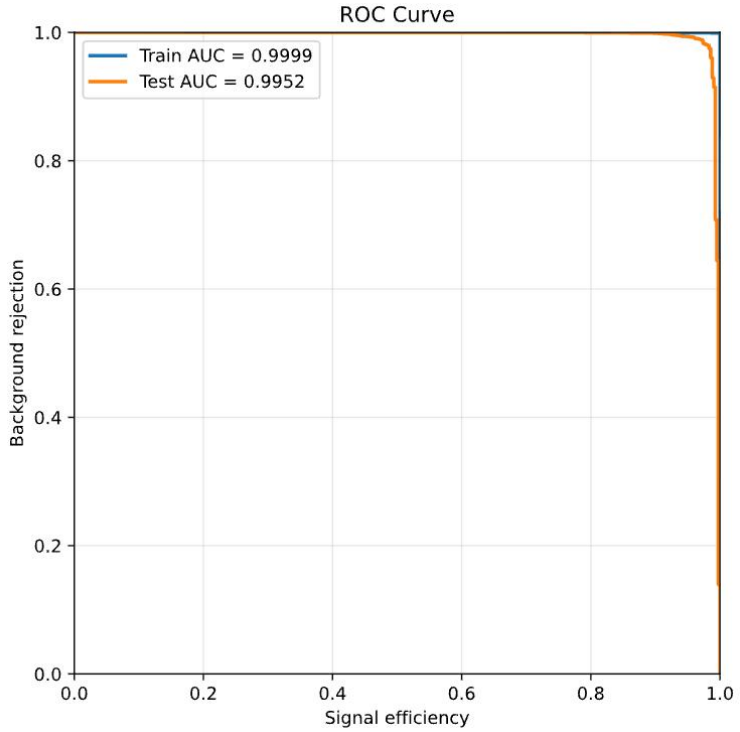
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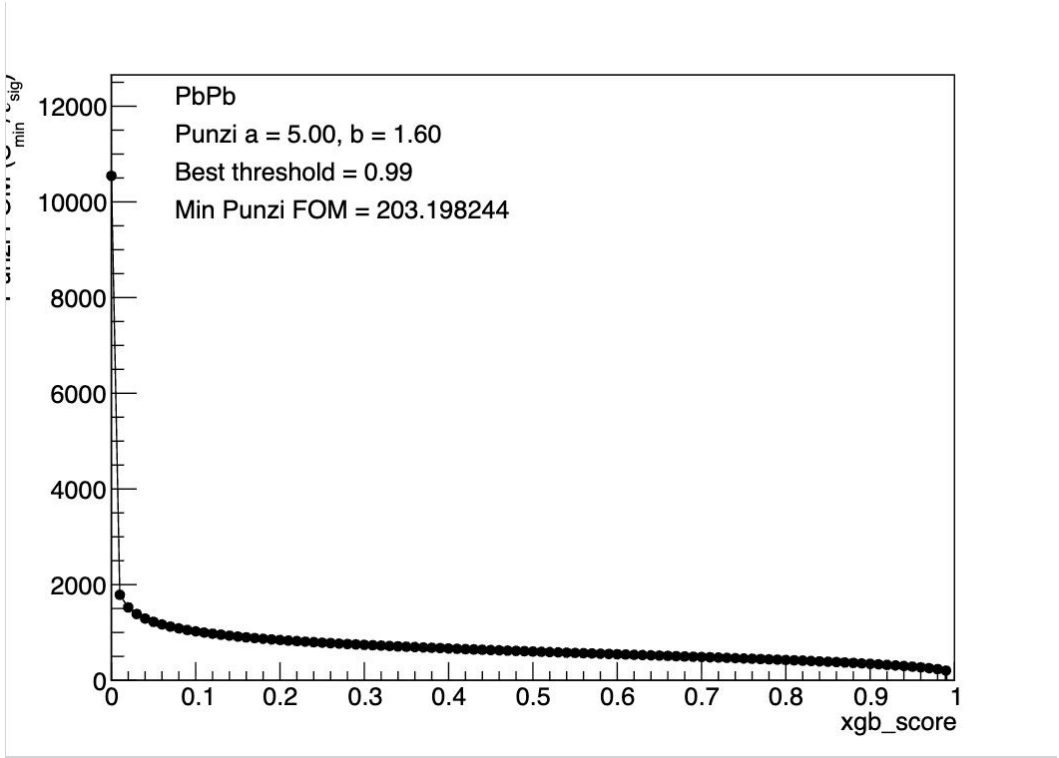
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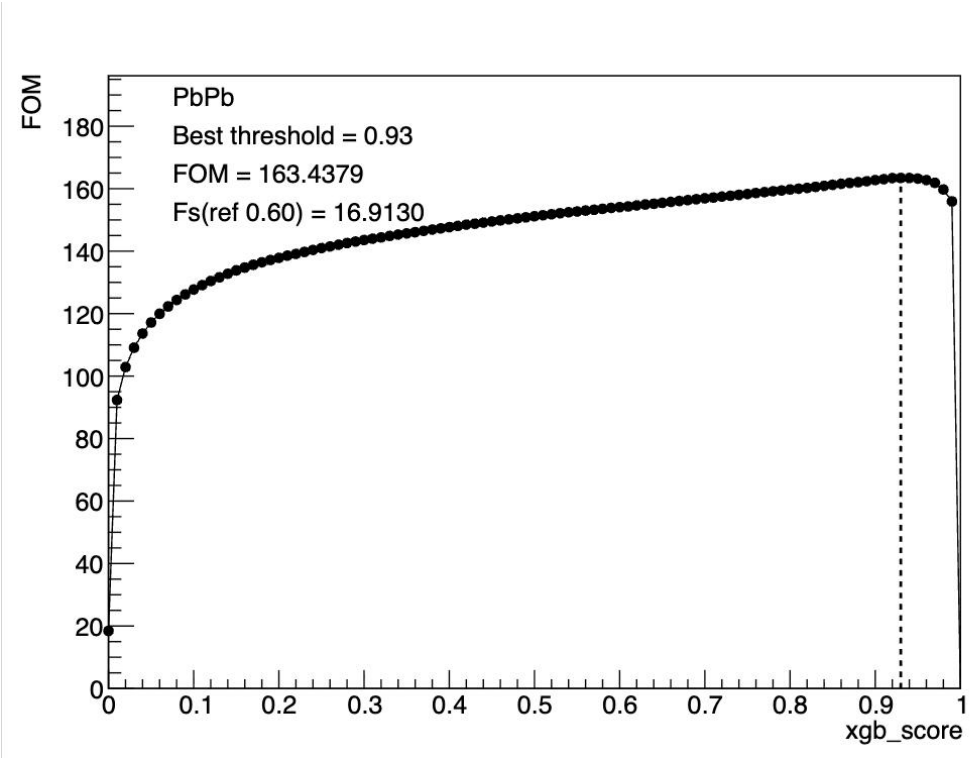




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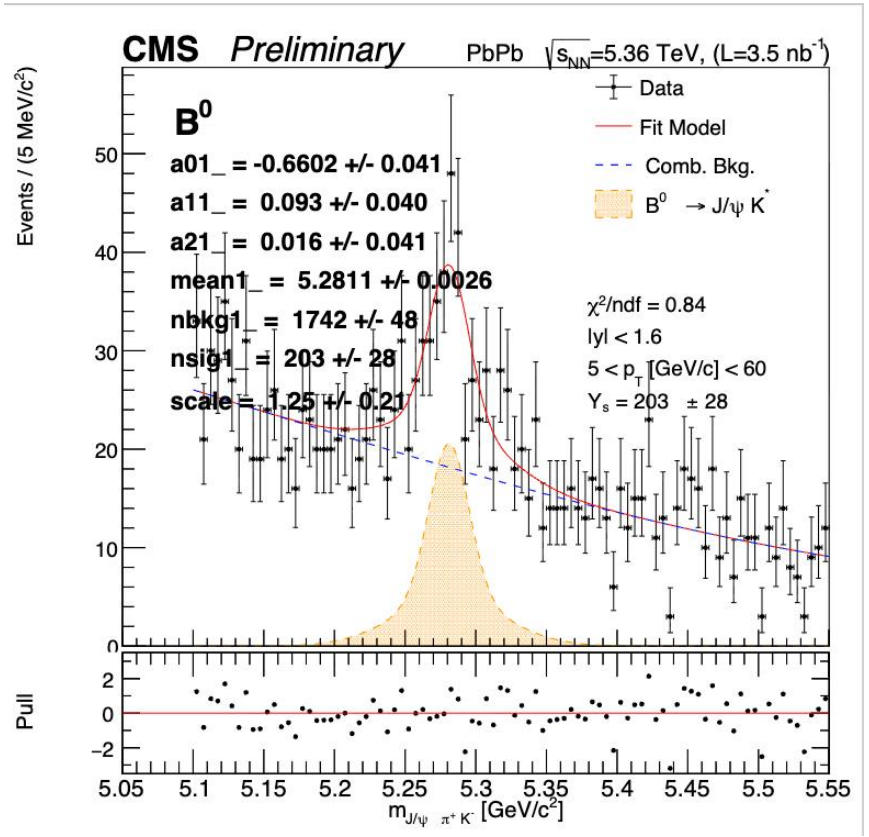


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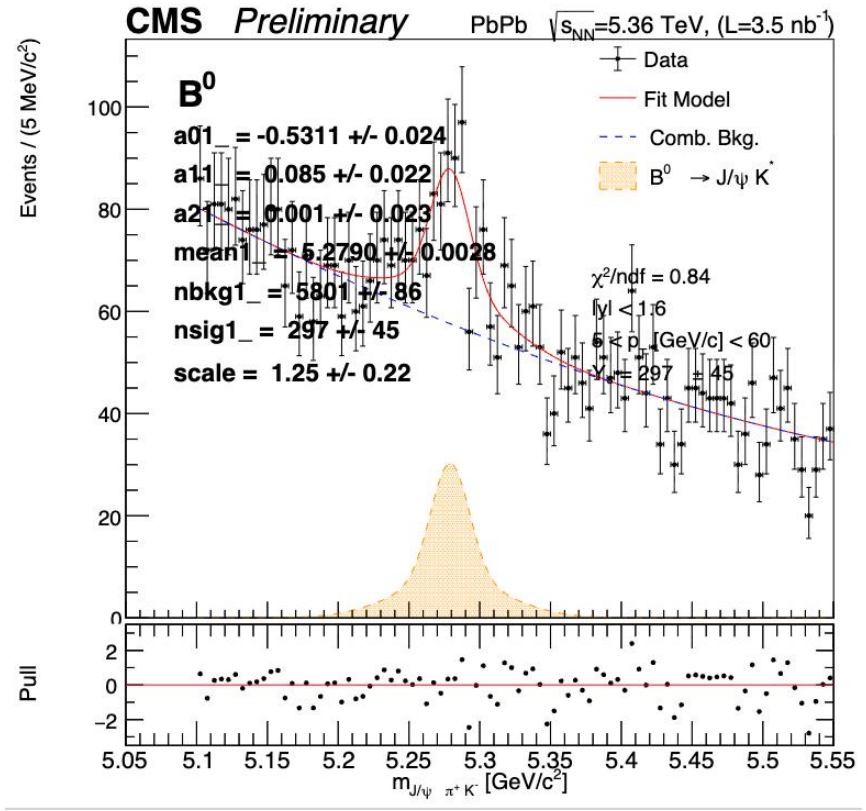


FOM

2024 pbpb : $B^0 \rightarrow JK^*$



with punzi optimal cut



with fom optimal cut

Thank you

BACKUP

Muon & Track Selection

Muons

❑ **Soft muons:**
- **normalized $\chi^2 \leq 1.8$**
- **Hits:**
  tracker layers ≥ 6
  pixel Layers ≥ 1
- **Displacement from vertex:**
  $dz < 35$ cm
  $dxy < 4$ cm

❑ **Acceptance region:**
- $p_T \geq 3.5 GeV$ & $|\eta| < 1.2$
- $p_T \geq (5.47 - 1.89 \times |\eta|)\ GeV$ & $1.2 \leq |\eta| < 2.1$
- $p_T \geq 1.5 GeV$ & $|\eta| < 2.4$

❑ **HLT matching:**
Path: *"HLT_PPRefL1DoubleMu0_v6"*
Filter: *"hltL1fL1sDoubleMu0L1Filtered0PPRef"*

Tracks

❑ **Quality:**
- High purity tracks
- $\sigma_{p_T}/p_T < 0.1$
- N_{hits} (pixel + tracker hits) ≥ 11
- $\frac{\chi^2}{ndf}/N_{hits} < 0.18$

❑ **Acceptance:**
- $p_T > 0.5$GeV
- $|\eta| < 2.4$

Candidates

❑ **Fiducial region:**
- $p_T > 5$GeV
- $|y| < 2.4$

Di-muon system

❑ Opposite muon charges
❑ Common vertex probability > 1%
❑ System's mass within 0.15 GeV from J/Ψ mass

Test between different punzi formula

minimize $\mathrm{FOM}_{\mathrm{punzi}} = \frac{S_{\min}}{\epsilon}$

If $a \approx b$, $S_{\min} = \frac{a}{2} + \sqrt{B}$

gaussian, $S_{\min} = b^2 + 2a\sqrt{B} + b\sqrt{b^2 + 4a\sqrt{B} + 4B}$

a more smooth version, $S_{\min} = \frac{a^2}{8} + \frac{9b^2}{13} + a\sqrt{B} + \frac{b}{2}\sqrt{b^2 + 4a\sqrt{B} + 4B}$

For cut optimization purposes, the result does not depend strongly on a and b, but given the usual exclusion at 95% and 5-σ level for discovery, a good choice can be a=2 and b=5.[1]

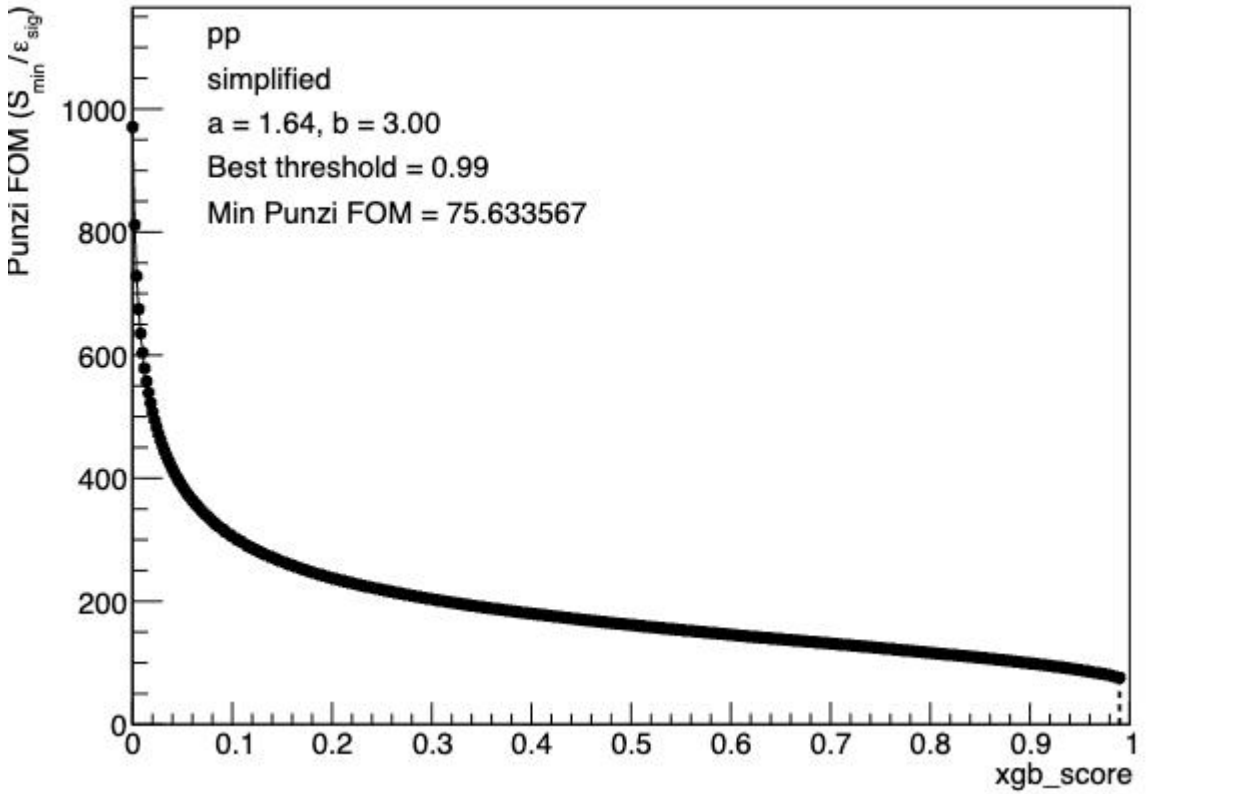
A test with different FOM (simplified, gaussian, improved) and (a,b) is done with Bu ppRef dataset and 12 variables (Btrk1dR, Btrk1Pt, Bcos_dtheta, Bnorm_svpvDistance_2D, Bchi2Prob, PVz, BujvProb, Btrk1Eta, Bmu1pt, Bmu2pt, Bmu2y, Bmu1y).

The conclusion is that punzi is stable between all the settings as claimed, but always leads to a tight cut.

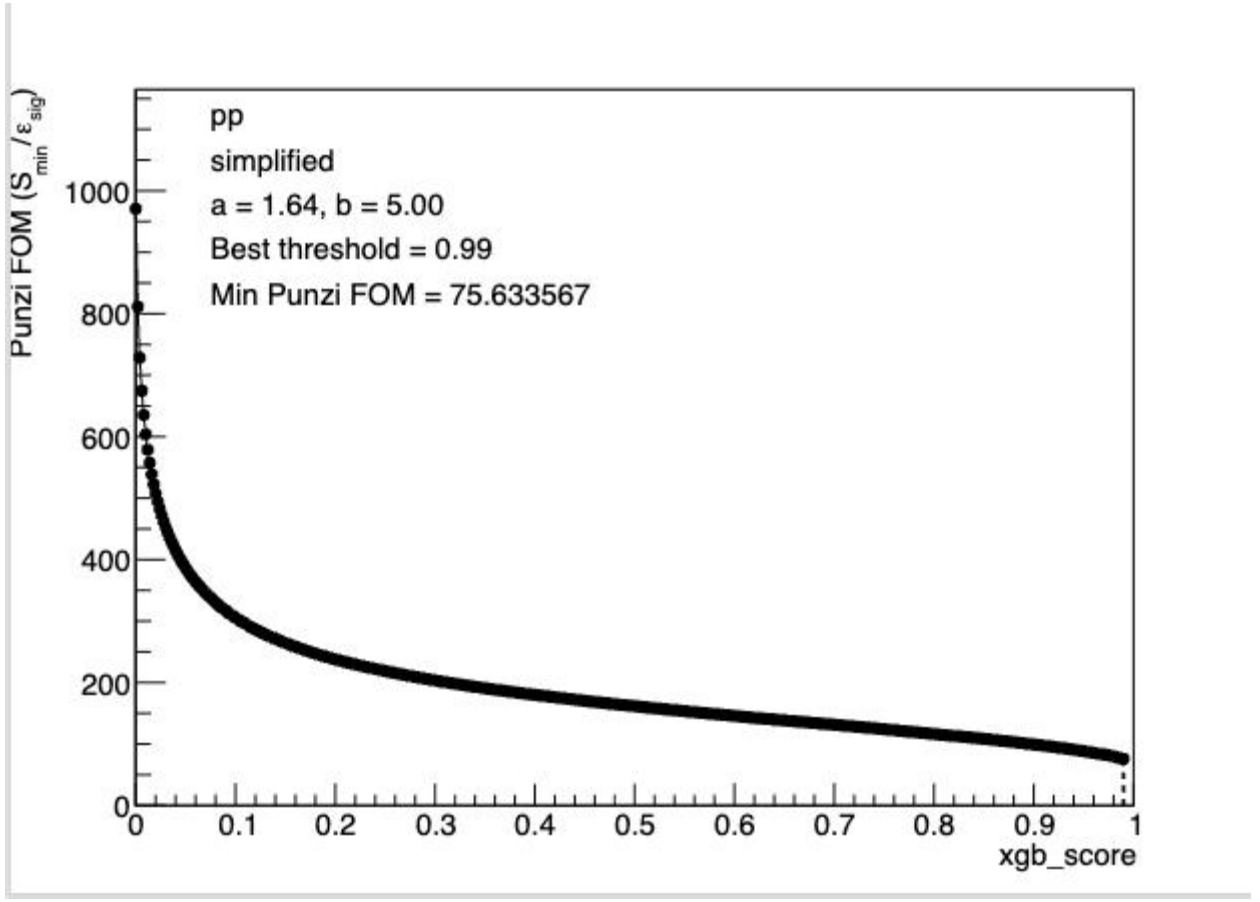
p.s. The punzi shown above is gaussian version with (a,b) = (5,1.6)

[1]https://twiki.cern.ch/twiki/bin/viewauth/CMS/PunziFom
[2]https://arxiv.org/abs/physics/0308063

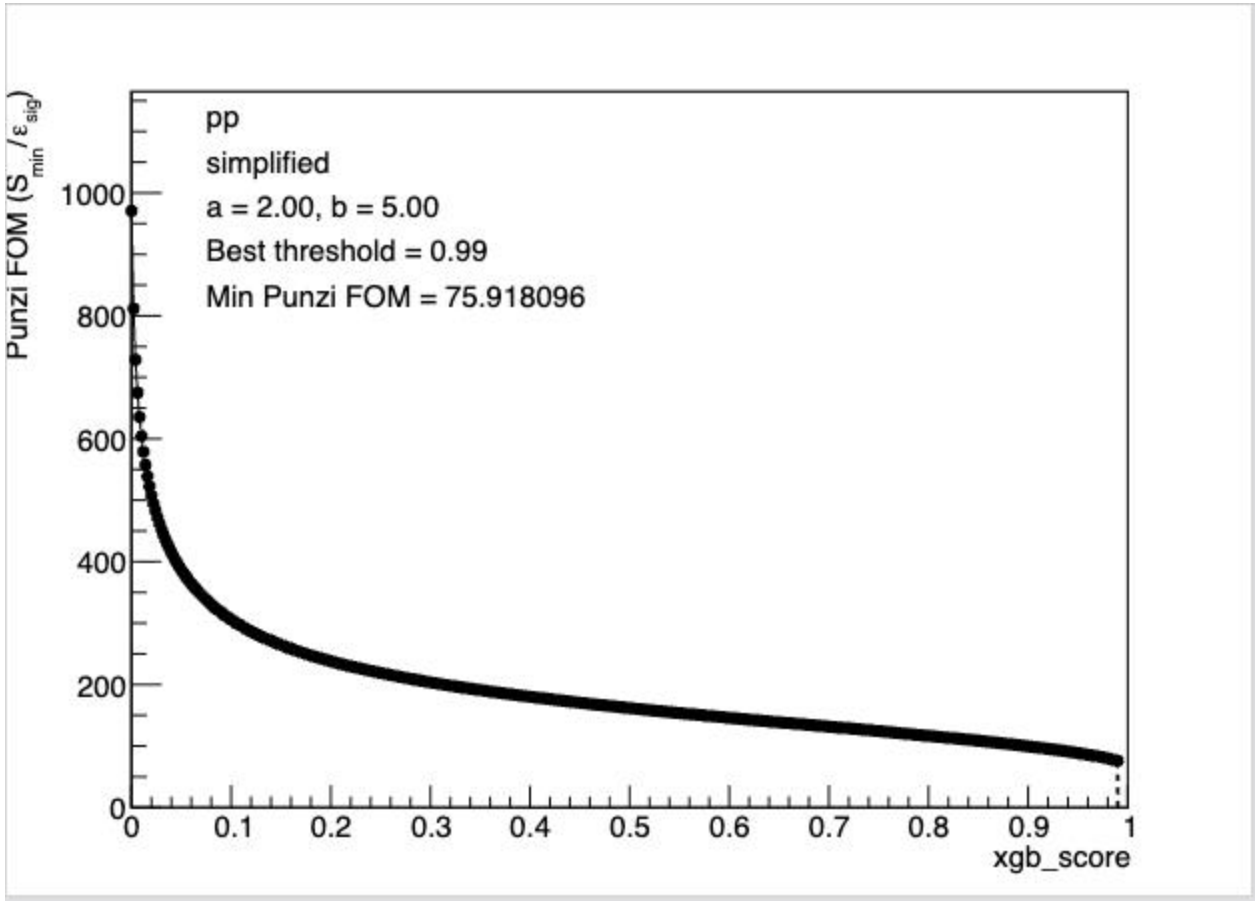
Bu_pp24_12variables_simplified-punzi_(a,b)=(1.64,3.0)



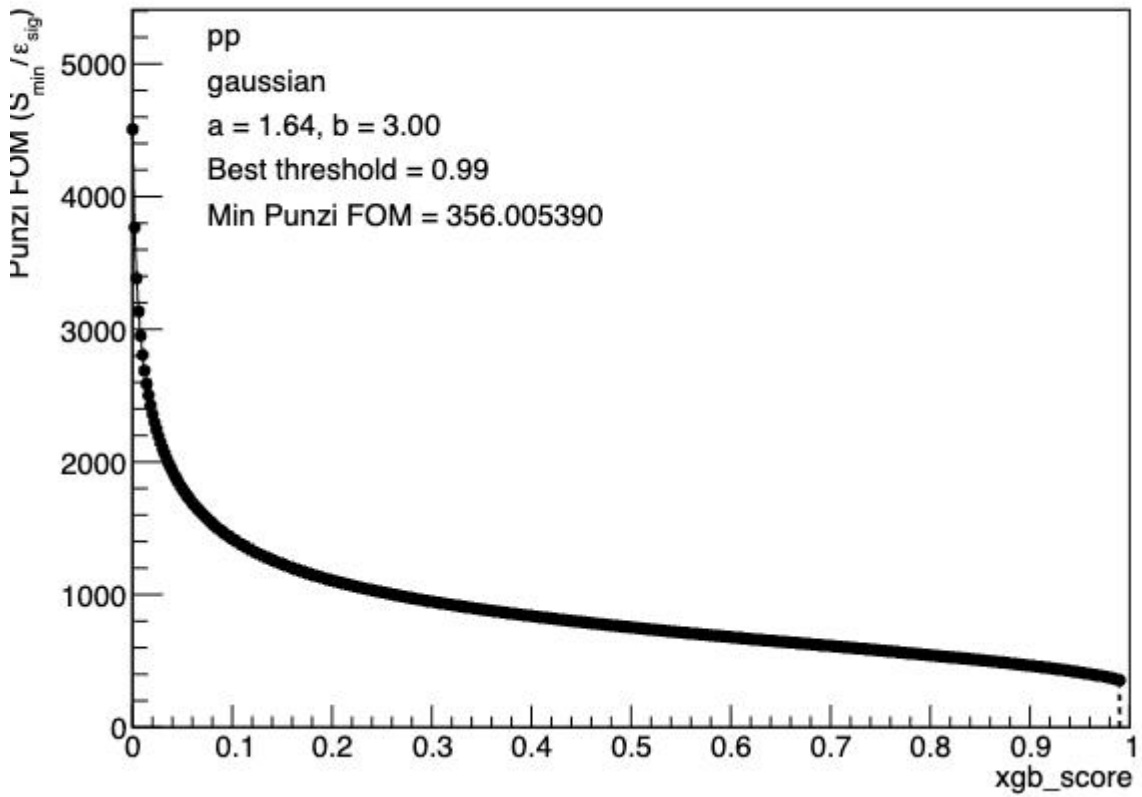
Bu_pp24_12variables_simplified-punzi_(a,b)=(1.64,5.0)



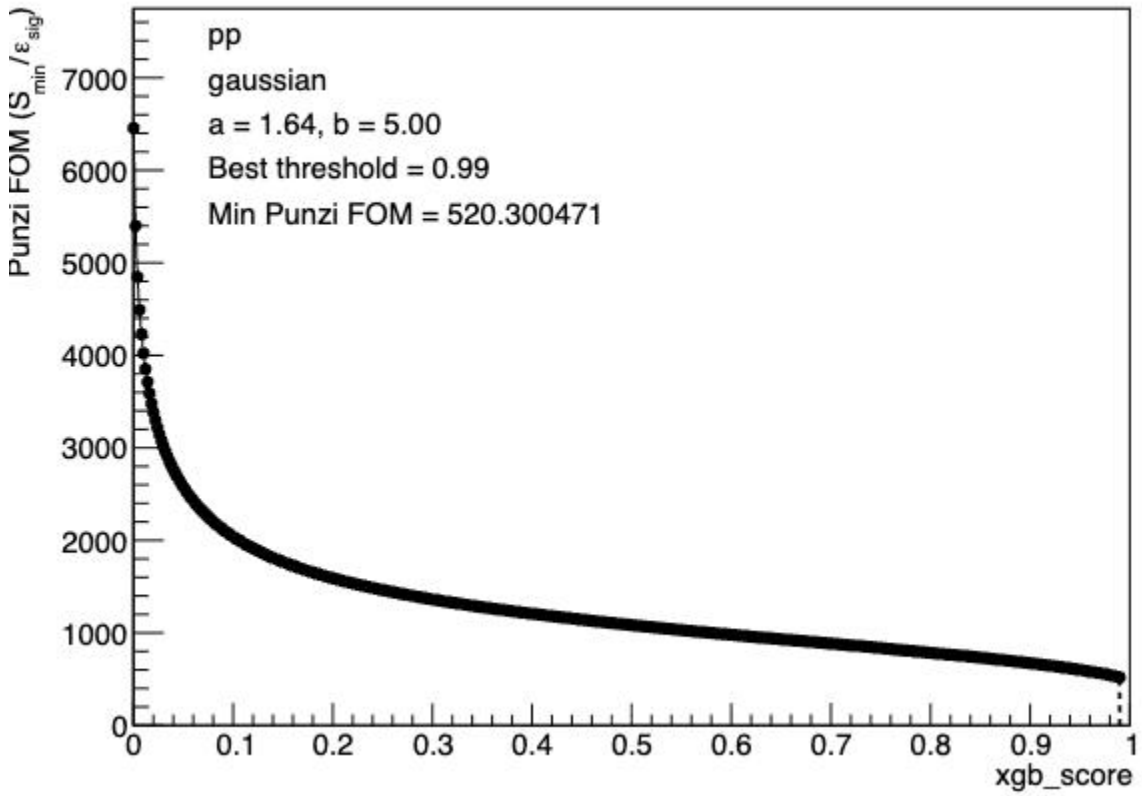
Bu_pp24_12variables_simplified-punzi_(a,b)=(2.0,5.0)



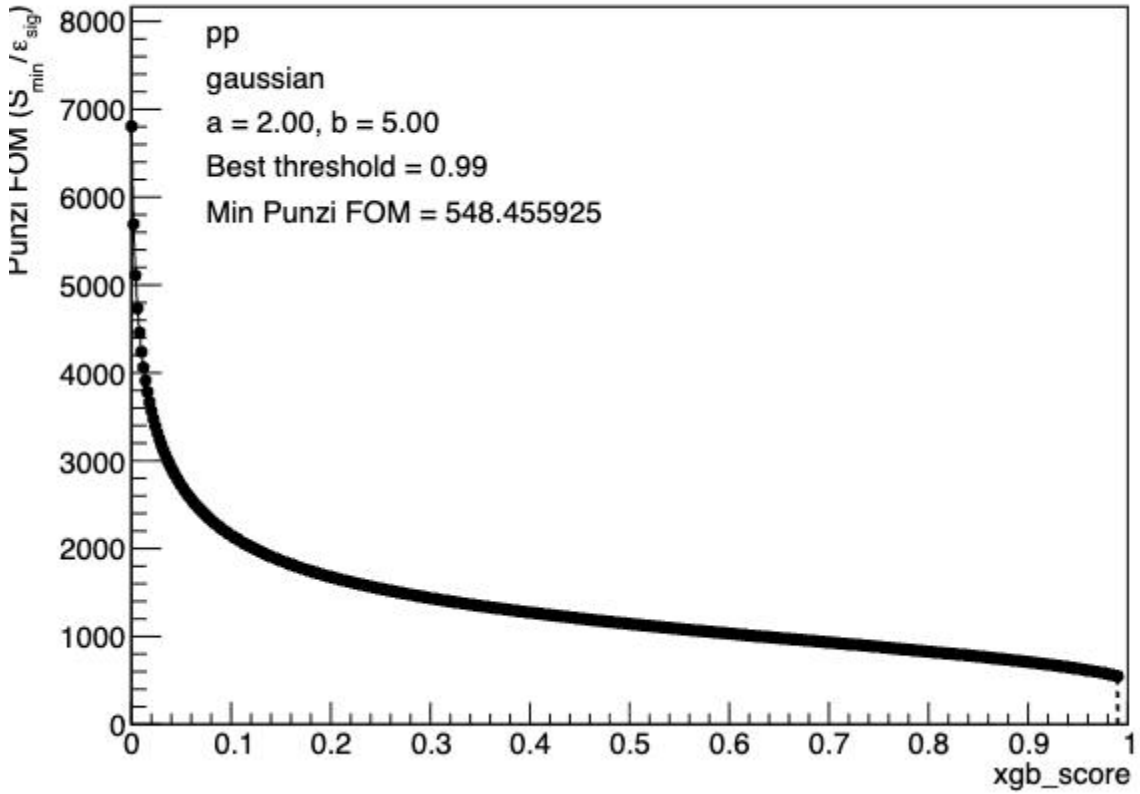
Bu_pp24_12variables_gaussian-punzi_(a,b)=(1.64,3.0)



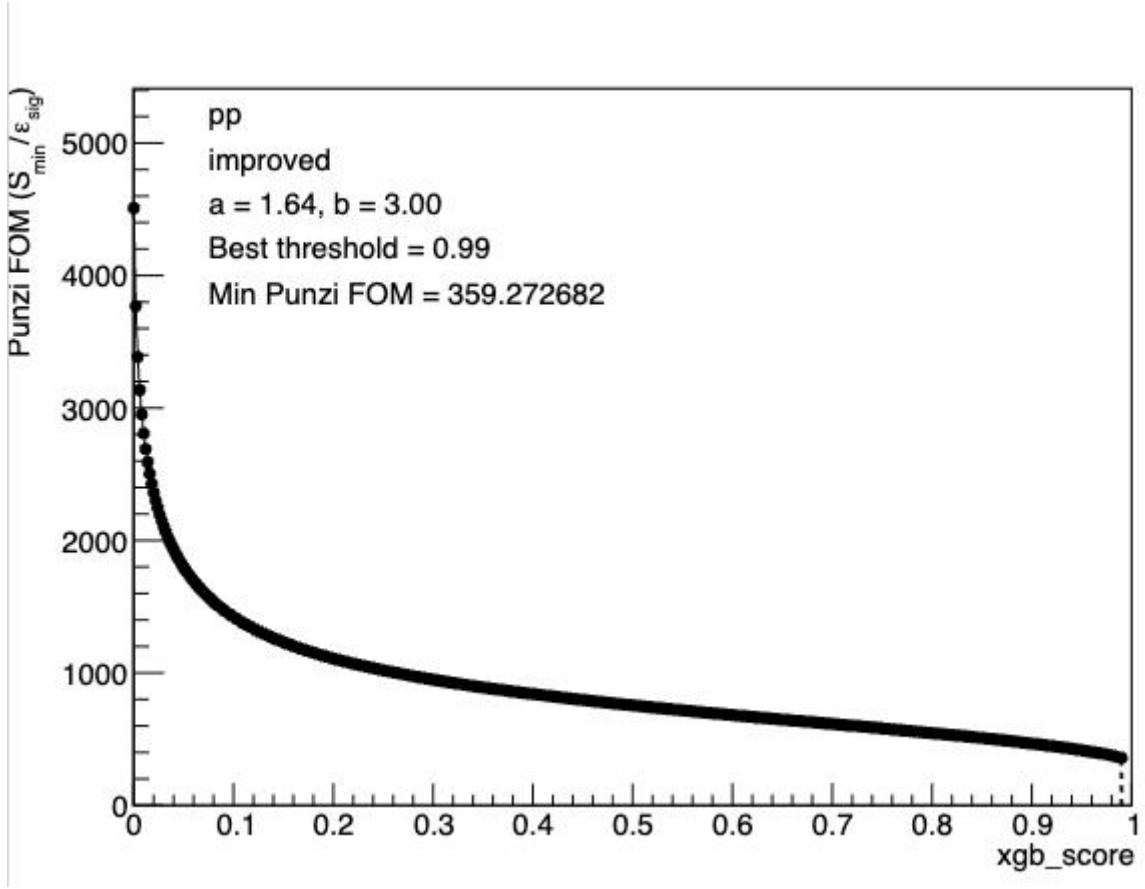
Bu_pp24_12variables_gaussian-punzi_(a,b)=(1.64,5.0)



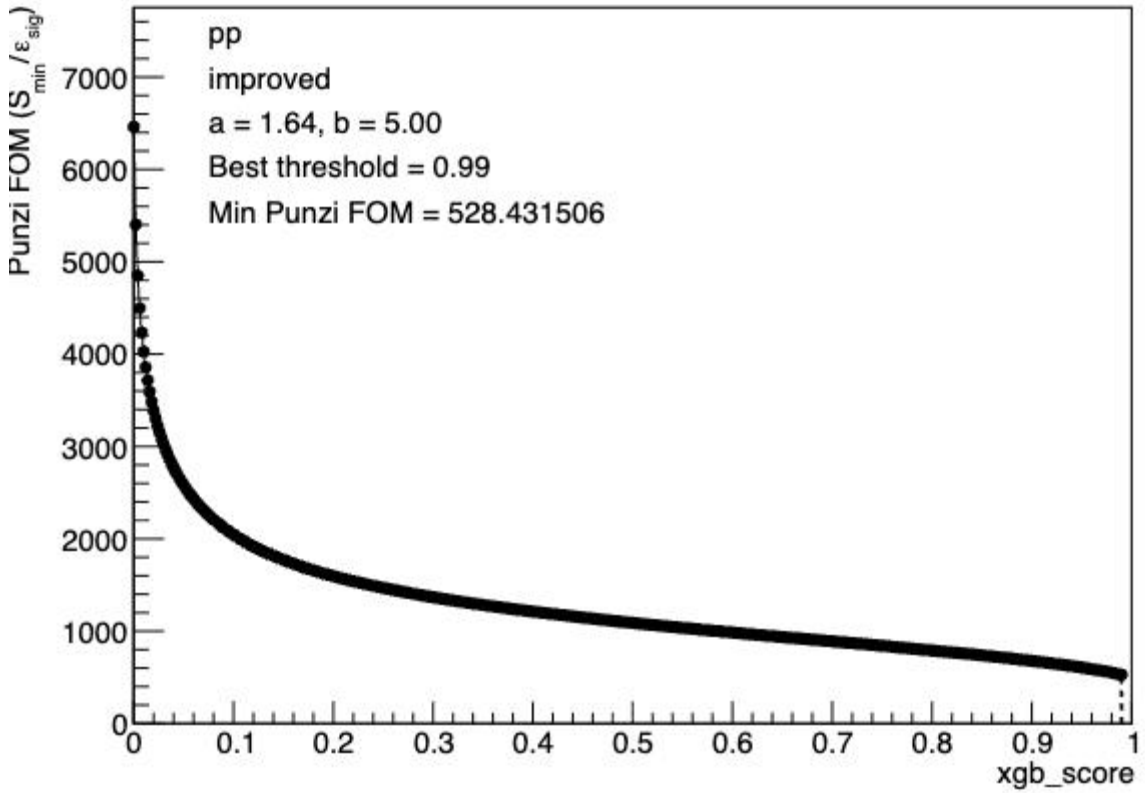
Bu_pp24_12variables_gaussian-punzi_(a,b)=(2.0,5.0)



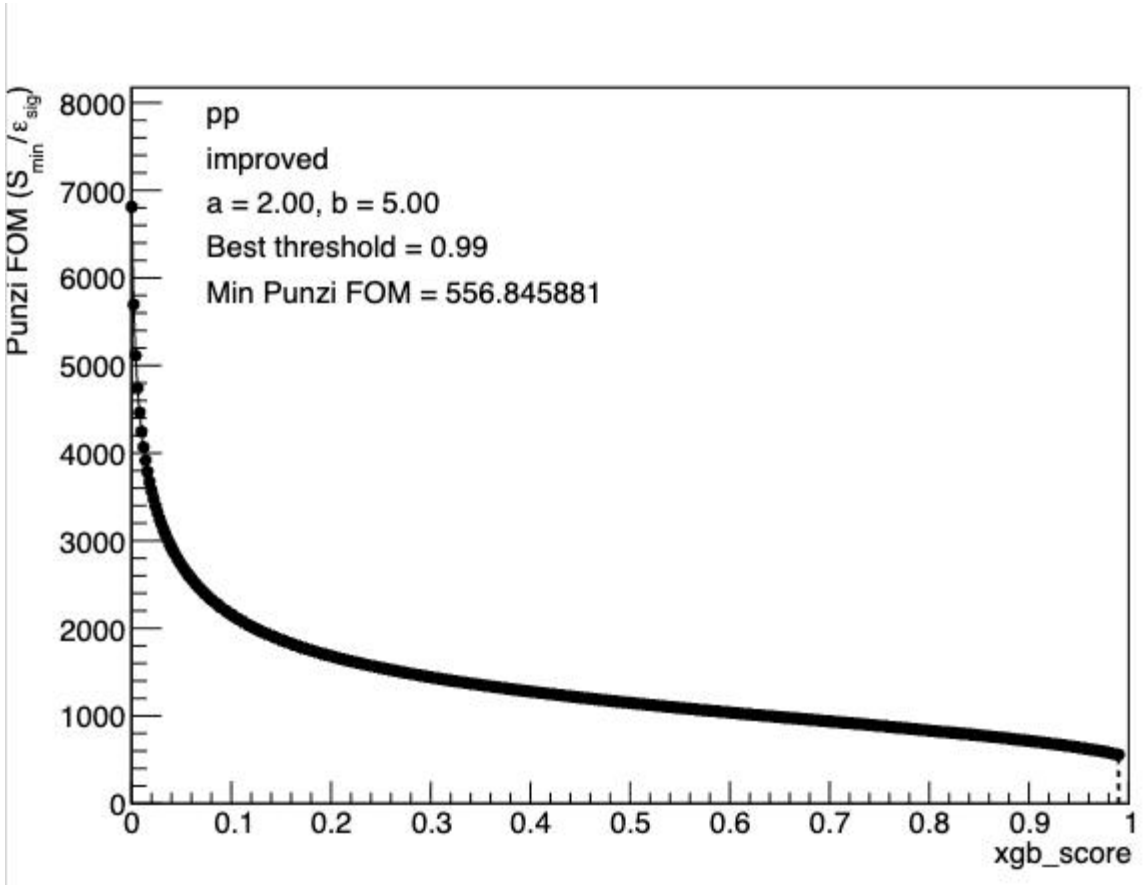
Bu_pp24_12variables_improved-punzi_(a,b)=(1.64,3.0)



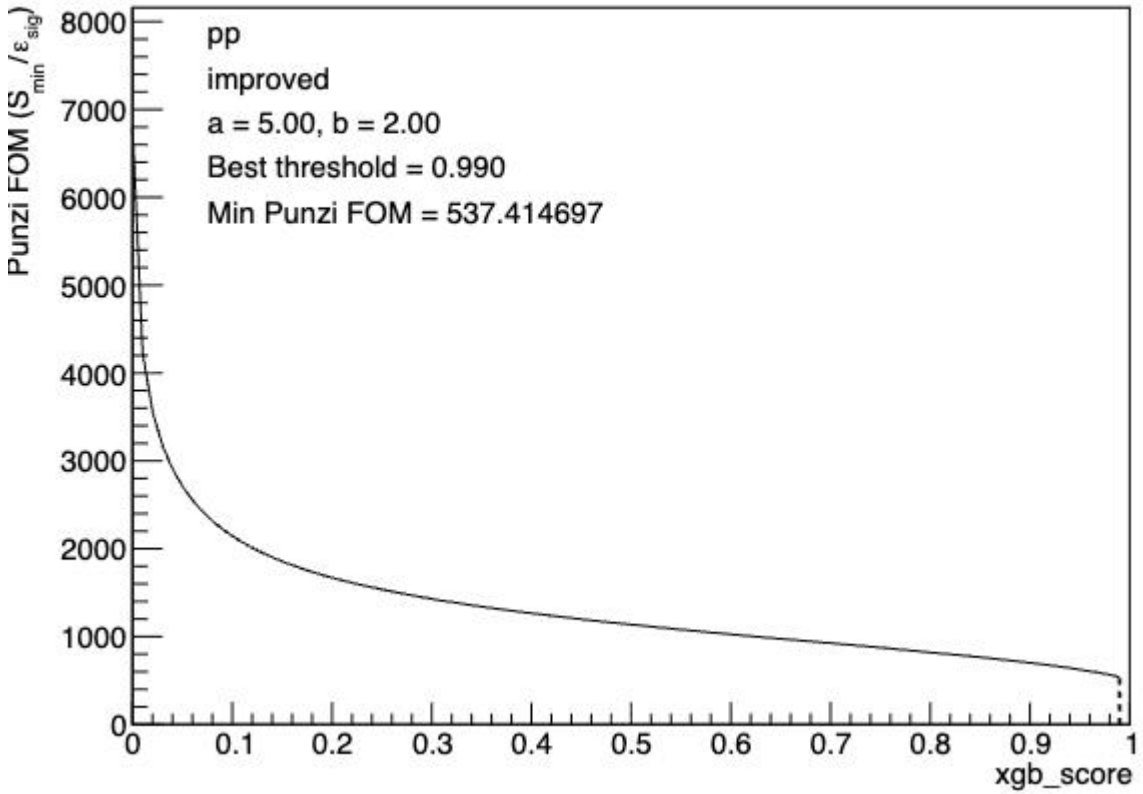
Bu_pp24_12variables_improved-punzi_(a,b)=(1.64,5.0)



Bu_pp24_12variables_improved-punzi_(a,b)=(2.0,5.0)

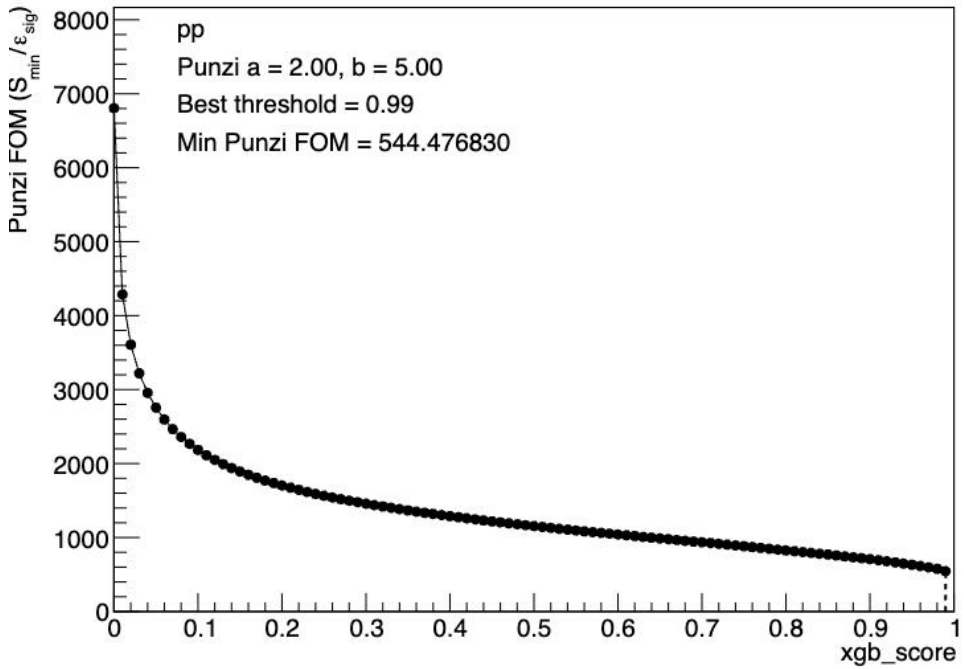


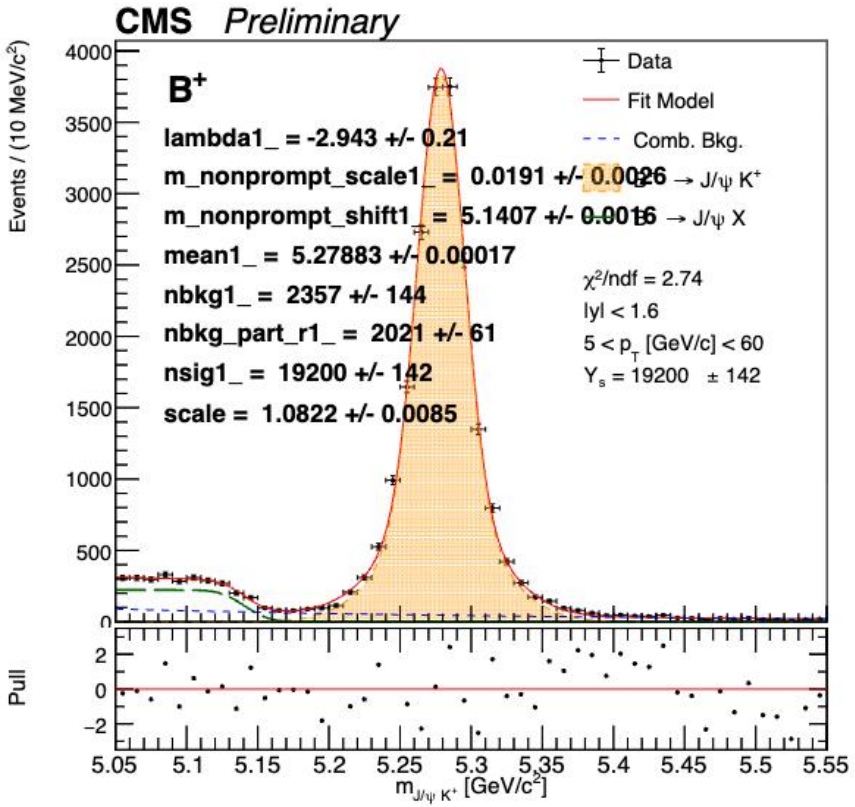
Bu_pp24_12variables_improved-punzi_(a,b)=(5.0,2.0)



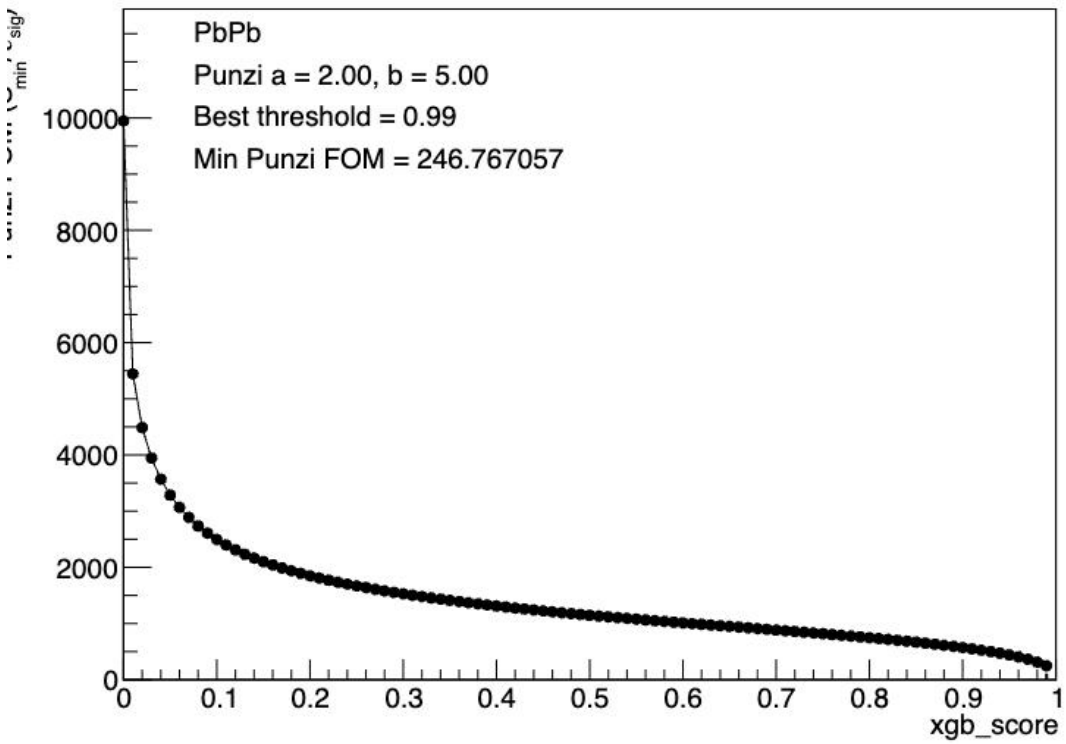
Bu_pp24_10variables_gaussian-punzi_(a,b)=(2.0,5.0)

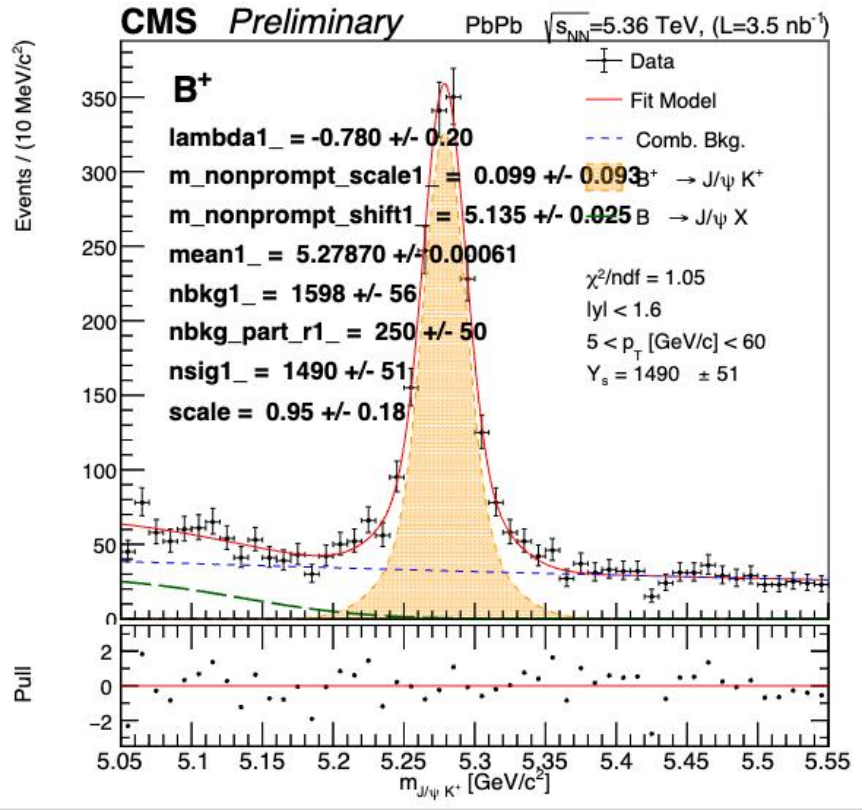
In an old version, (a,b) = (2.0,5.0) from CMS twiki was used,
which is actually opposite with the true definition



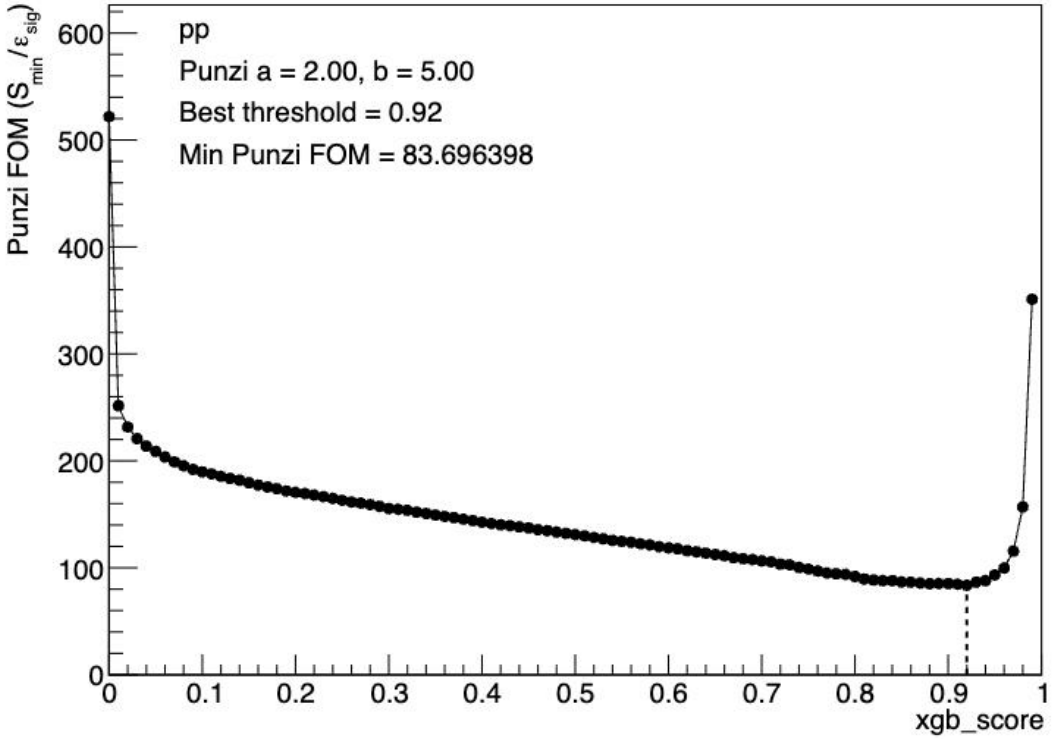


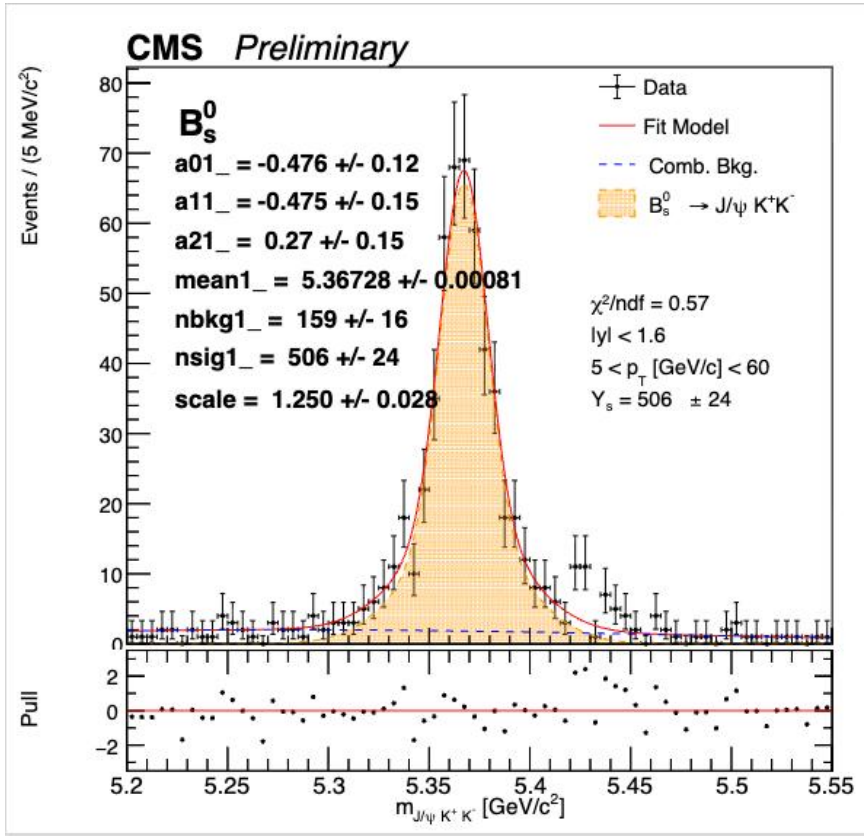
Bu_pb24_10variables_gaussian-punzi_(a,b)=(2.0,5.0)





Bs_pp24_14variables_gaussian-punzi_(a,b)=(2.0,5.0)





Bs_pb24_14variables_gaussian-punzi_(a,b)=(2.0,5.0)

